

Optimization in offshore wind farms

– a survey of problems, models, and solution approaches

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Abstract

Offshore wind offers clean and renewable energy on a large scale, and hence plays a central role in the green energy transition. This paper provides a digested survey of optimization problems, models, and solution approaches related to offshore wind farms that gives the reader an introduction to this exciting and growing research area. The paper is structured according to the five phases of the offshore wind farm life cycle: *Area Selection*, *Wind Farm Design*, *Installation*, *Operations and Maintenance*, and *End of Life*. For each phase, we identify relevant Operations Research problems, and for the most mature problems, for which a substantial body of literature exists, we present mathematical formulations. For these problems we also discuss available data sets and provide an overview of solution methods. Furthermore, a structured classification of the relevant literature within each phase of the life cycle is provided in summary tables. Finally, we highlight future trends and discuss future research directions along several axes before concluding.

Keywords: Offshore wind farms, Facility location, Network design, Vessel routing and scheduling, Fleet size and mix

1. Introduction

Offshore wind farms (OWFs) play a significant role in supplying the world with renewable energy due to their vast energy potential and complementarity with other renewable energy sources, such as solar energy and biogas. During the past decade, global offshore wind energy capacity has increased more than tenfold, and the Global Wind Energy Council (GWEC) projects that annual installations will continue to grow exponentially throughout this decade and beyond (see Figure 1). Europe and China currently account for 95% of the offshore wind market (GWEC, 2025). This is directly linked to ambitious goals on energy transition in both regions (European Commission, 2019; State Council of the People’s Republic of China, 2021), in which installing substantially more offshore wind energy is one of the cornerstones. Both Europe and China are expected to continue their dominance within offshore wind growth in the coming years, while other markets in Asia-Pacific, such as India, Japan and Vietnam, as well as the US market, are expected to increase their market shares. On a longer time scale, countries like Brazil, Colombia, Australia and Canada are also predicted to increase investments in OWF (GWEC, 2025).

Each OWF consists of a number of *components*, with the most central being the *offshore wind turbines* (OWTs) that harvest wind energy and convert it to electricity. Smaller OWTs produce less than 1 MW, while the largest may reach a nominal production of 16–22 MW. The three-bladed *horizontal axis wind turbine* (HAWT) design (see Figure 2a) is by far the most widespread offshore, while the *vertical axis wind turbine* (VAWT) design (see Figure 2b) is currently limited to pilot projects. Therefore, unless explicitly stated otherwise, when referring to an OWT, we understand that it is a HAWT. The HAWT consists of three *blades* that are attached through the *hub* to the *nacelle*, housing the *gearbox* and the *generator*. The nacelle is placed on top of the *tower*, and the entire structure is supported by a *foundation*. The VAWT, on the other hand, has two (or more) blades that rotate around the axis of the tower. The gearbox is placed inside the bottom of the tower, with the generator next to it, on the foundation.

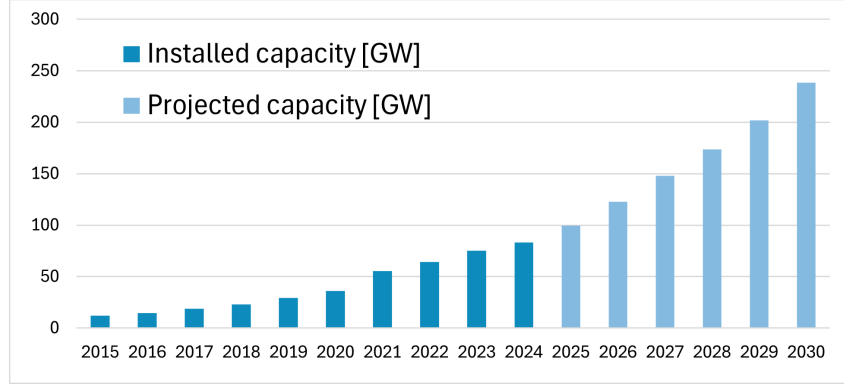
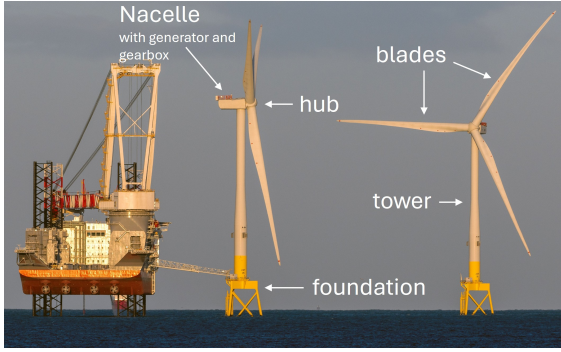
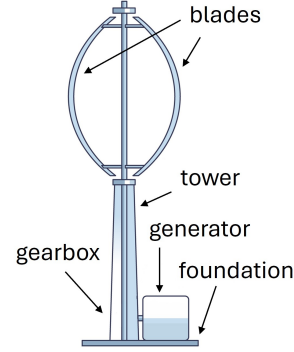


Figure 1: Global installed and projected offshore wind energy capacity from 2015 to 2028 in GW, based on Global Wind Energy Council’s (GWEC’s) annual reports.

The produced electrical energy is collected through a network of *cables* called the *inter-array network* that connects the turbines to one or more substations. The *substation* transforms the power to a higher voltage, so that it can be transmitted onshore through one or more *export cables*.



(a) Illustration of HAWTs, based on photo of Aberdeen Bay Wind Farm by Gannet77/iStock.



(b) Illustration of a VAWT, based upon illustration by VectorMine/iStock.

Figure 2: Illustration of OWT components in different OWT designs.

In comparison with their onshore counterparts, which are more restricted in terms of locations, altitudes, and *wake effects* (wind shadows) from surrounding mountains and vegetation, OWFs are very attractive due to the higher and steadier wind speeds at sea together with more available space. However, placing wind turbines (WT) in wind farms (WFs) offshore also poses challenges in terms of the harsh maritime environment that wears on the wind turbines and makes the sites far less accessible. Adding to the complexity, many sites close to shore are already occupied. Consequently, new OWFs are placed farther away from the shore, requiring longer export cables and resulting in more difficult logistics operations that require highly specialized vessels (IRENA, 2025). The increased water depth might also mean that *floating* foundations are necessary, which introduces new challenges (IRENA, 2023).

Due to these challenges, the *levelized cost of electricity* (LCOE) for offshore wind in 2024 was 0.073 €/kW, compared to just 0.031 €/kW for onshore wind (which was below the LCOE for the cheapest fossil fuel-fired electricity option) (IRENA, 2025). Within the energy sector, LCOE is used as a metric to compare the cost-efficiency of electricity production across different energy sources. For a given source it is defined as the *net present value* (NPV) of the *electricity production cost*, divided by the *electricity production*, over its lifetime. The total cost of electricity production consists of two components: The long-term *capital expenditures* (CAPEX) resulting from long-term investments, and the short-term *operational expenditures* (OPEX) resulting from day-to-day operations. In a typical OWF, CAPEX accounts for approximately 72% of the cost, while OPEX accounts for the remaining 28% (BVG Associates, 2019).

Currently, there is a tendency toward building larger and larger OWTs such that the costs are averaged

over greater electricity outputs (IRENA, 2024). However, there is great potential for cost-savings in more efficient planning of resources throughout the *life cycle* of OWFs. This leads to a number of interesting and challenging optimization problems. So far, a large part of the research has been driven by engineers with technical knowledge, e.g., on fluid dynamics, or mechanical and structural design. However, in recent years, research has moved into the Operations Research (OR) society, where a large toolbox of techniques can help to significantly reduce the LCOE of offshore wind.

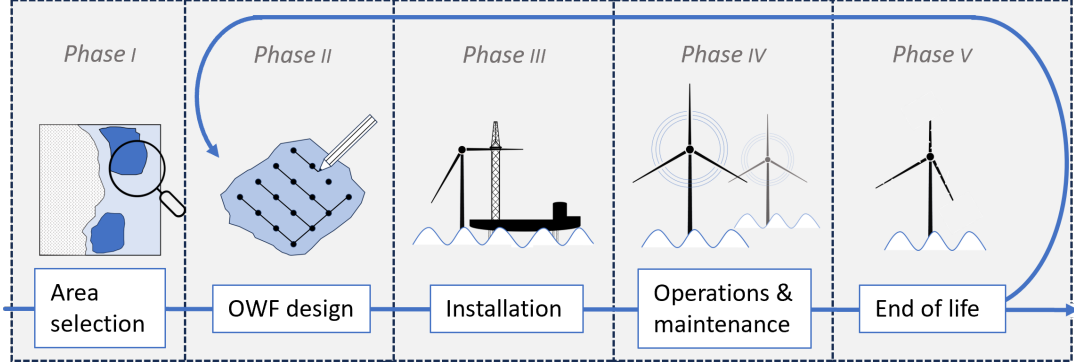


Figure 3: Life cycle of an offshore wind farm.

The OWF life cycle can be divided into five phases (see Figure 3) beginning with phase I, *area selection*, where the site of the OWF is chosen. That is followed by *OWF design* in phase II, covering both the location of OWTs and designing the cable network. Next, phase III focuses on the actual *installation* of the OWF, including all the OWTs, cables, and substation(s). With an OWT *design life* of 25–30 years, phase IV, *operations and maintenance*, has the longest duration. It entails both performance optimization of the OWF and the planning and execution of maintenance operations. Lastly, in phase V the OWF reaches the *end of life*. Depending on the state of the OWTs, the OWF might continue to harvest energy for an extended period before it is eventually decommissioned in a process reverse to that of the installation. After phase V, a new OWF life cycle in the same area might start with phase II.

The **purpose** of this survey is to give a curated overview of the optimization problems occurring throughout the OWF life cycle with the aim of reducing LCOE, and to present mixed integer programs (MIPs) for a selection of these. The problems discussed, and models presented, are high-level models that do not require detailed knowledge in engineering topics like fluid dynamics and mechanics. Instead, we focus on giving the readers — researchers as well as practitioners — an introduction to this exciting and growing research area that is new to many in the OR community.

The main **contributions** of this article may be summarized as follows: i) For each phase of the OWF life cycle, we outline the applied and historical context to establish the problem settings and then survey the relevant literature. ii) The surveyed literature within each phase is classified in tables according to various problem-specific criteria. iii) We identify gaps in the literature, highlight trends within offshore wind energy, and consider promising avenues for future research.

Due to the different levels of maturity of the literature concerning each phase, the level of detail provided with respect to problem definitions, models and solution approaches, differ. The OWF design phase has by far the most mature literature, with several classes of standardized problems, and a rich body of solution approaches, covered in this survey. The installation and the operations and maintenance phases also contain a substantial, though less developed, body of literature, with current research focusing primarily on proposing richer and more realistic problems, and formulating them as mathematical models. For the area selection and end of life phases, only limited literature is available, and therefore only a brief discussion of possible problem settings is provided for these phases.

We use the OWF life cycle as the basis for the **organization** of the remainder of this survey. We begin in Section 2 with area selection (Phase I), followed by OWF design (Phase II) in Section 3. Next, in Section 4, the installation (Phase III) is studied; Section 5 considers operations and maintenance (Phase IV), and Section 6 focuses on the end of life (Phase V). Future trends and research directions are discussed in Section 7 before we conclude in Section 8.

2. Area selection

In most European countries, the establishment of OWFs is based on a tender. The government selects an area of sea for energy production and specifies how much energy (in MW) should be produced. The target energy production is typically limited by how much energy the electrical grid onshore can support, or by the wind energy available within the given area. The company with the best proposal with respect to cost and score on various parameters wins the tender, and is allowed to build and operate the OWF for a specific amount of time (typically 20/25 years). The tender may take into account how good the proposal is at utilizing the area (electricity production per square kilometer), how it affects its surroundings, as well as environmental aspects.

However, in recent years, countries have also opened up for companies establishing OWFs on their own initiative. Denmark had an “open door scheme” for renewable energy installations at sea in 1999–2023, and in the UK, several larger areas have been made available for OWFs. Projects are evaluated according to their area usage, shape of the selected area, and energy production. In both cases, it opens up for the *area selection problem*. Formally speaking, from a developer perspective, the task is to select an area subject to some regulatory constraints, such that the LCOE is minimized. The inputs are the available area(s) for locating the farm, a map of the seabed (depth, soil conditions, and obstacles), information about the wind speed at each position, and some regulatory constraints. The profitability of an OWF depends on the wind speed (more wind – more energy), seabed (deeper sea – bulkier foundations), and distance from shore (longer distance – more expensive cables). Furthermore, existing upstream OWFs can reduce wind speed through wake effects, making the distance between OWFs a factor worth considering (NYSERDA, 2018). In order to make a good area selection, it is necessary to estimate the costs of the offshore wind farm layout and to estimate the expected production. However, it is often sufficient with a brief estimate. Once a good area has been selected, a more detailed analysis can be carried out.

Fischetti and Fraccaro (2019) used a combination of machine learning and optimization to solve the area selection problem. A machine learning algorithm was trained on more than 3000 optimized layouts of real-world farms, considering rectangular shapes of the area, and assuming that turbines are placed on a regular grid. To train the model, a set of features were selected. Once trained, the machine learning algorithm could scan a large number of possible locations to predict the expected production. Generally, there was a very good correlation between predicted production and true production (found by solving the corresponding optimization problem).

Feng and Shen (2020) proposed a two-stage heuristic to decide the shape, orientation, and layout of an OWF. To simplify the problem, the boundaries are assumed to have a parallelogram shape. The first-stage heuristic is used to select good shapes and orientations, and the second-stage algorithm optimizes the layout of the turbines for each selected shape and orientation. Cazzaro et al. (2022), on the other hand, proposed an algorithm consisting of three steps: (i) a macro-scale algorithm selecting the approximate location of the wind farm out of larger regions, (ii) a meso-scale algorithm generating the optimal shape of the OWF, and finally (iii) a micro-scale optimization choosing the best location of the individual turbines within the given area.

3. Offshore wind farm design

The OWF layout problem consists of two intertwined subproblems: the wind farm layout problem, where the task is to place turbines within a given area such that the energy production is maximized and building costs are minimized, and the wind farm cable routing problem, where the task is to connect all turbines to a base station at minimal cost.

The layout of an OWF is perhaps the most studied step in the life cycle of OWFs due to the huge installation costs and the impact on the expected energy production. Fischetti et al. (2020) showed that a good layout of an OWF can increase earnings by a significant amount (e.g., 12.6 M€ for *Borssele 1*). The problem consists of several sub-tasks: selecting the correct wind turbine, locating turbines and substation(s) within the available area to maximize energy production, cable selection, and finally cable routing. Fischetti and Pisinger (2019) give a good introduction to the various optimization problems in this phase. First, an area is selected and the location of the substations is determined. Next, the turbines are placed to maximize production, and then they are connected with cables.

We assume that we have as input a wind rose describing the wind profile and a map of the seabed. The *wind rose* represents historic or forecasted wind information, while the *seabed map* defines which positions are allowed to be used, obstacles on the seabed that need to be avoided, as well as water depths (making it possible to calculate the foundation costs at every location). In the *homogeneous problem*, it is assumed that all turbines are identical, and have the same energy production curve. The homogeneous problem is the most studied, and also in practice by far the most applied design, since installation and maintenance costs are lower when only a single turbine type is used. In the following we will therefore only consider the homogeneous version. A survey of *non-homogeneous problems* can be found in Tao et al. (2020b).

The *turbine selection problem*, selecting the right turbine for a given site, is an important problem. However, with the development of efficient algorithms for OWF layout, it is easy to try the (relatively few) turbine types, evaluate each configuration, and choose the best option. We will therefore not pursue this topic further. Typically, a layout problem has a *target electricity production* (in MW). Knowing the target production and the nominal production of each turbine directly gives the number of turbines T needed.

We start this section by considering the WF layout problem (WFLO) in Section 3.1, and then we consider the cable routing problem (WFCR) in Section 3.2. Finally, we consider integrated models for both layout and cabling in Section 3.3. The section is concluded with a discussion about stochastic decision making in Section 3.4, and pointers to public data sets in Section 3.5.

Although we in the following focus on the offshore version of the problems, onshore problems can be handled in a similar way. Offshore problems are generally two-dimensional, while onshore problems often are three-dimensional where one has to take into account the wake effect from surrounding mountains, buildings, etc. Also, the cabling problem is different. In offshore wind farms the cables can be laid as straight lines, while onshore problems must avoid many obstacles. However, most of the differences can be handled by a more detailed pre-processing in the onshore case.

3.1. Wind farm layout

The *wind farm layout problem* (WFLO), also referred to as *micro siting*, consists of placing T wind turbines within a given area $\mathcal{R}$ to maximize electricity production and minimize building costs. The area $\mathcal{R}$ may contain obstacles where it is forbidden to place turbines. As input we have one specific turbine type with specific characteristics, including the cost, and the nominal electricity production at various wind speeds.

We assume that there is only one *substation* where all energy needs to be collected. From the base station there is an *export cable* onshore. The dimension of the export cable is given by the target production. We also assume that the location of the substation is fixed, although various methods have been proposed to determine the number and location(s) of substations. A minimum distance $d^{\min}$ between turbines must be respected to avoid damage or excess tension. It is common to assume that the turbines should be five times the diameter of the rotor. (This decreases the fatigue degradation of WTs.)

Kaminsky et al. (1987) were the first to study the WFLO for a set of turbines installed *in line*. Mosetti et al. (1994) presented the first real WFLO in two dimensions, where turbines are to be placed on a 10×10 *grid*. The first algorithms to solve large-sized problems were proposed by Chen et al. (2013), followed by Fischetti et al. (2016) and Cazzaro et al. (2022). For recent surveys of WFLO, see Samorani (2013), Fischetti and Pisinger (2019), and Hou et al. (2019), or the very detailed literature overview by Cazzaro and Pisinger (2022b).

The power production of a turbine is highly affected by the wake effect. A *wake* is a region of lower wind speed that each turbine creates behind its rotor, which reduces the energy yield of downwind turbines. One can say that the turbine casts a *shadow* for downwind turbines. Various wake models have been proposed in the literature that make it possible to estimate the wake effect without the need for a Computational Fluid Dynamics simulation. One of the most widespread formulations is the *Jensen* model (Jensen, 1983), which assumes that the wake downstream of a turbine can be described by a trapezoid. Due to its simplicity and robustness, the Jensen model has been applied extensively to model wake losses, and several studies have shown that the calculated wake effect (for a complete WF) in practice matches quite well with the observed value. Several improved wake models have been proposed, including the *Ainslie* model, the *Larsen* model, the *Frandsen* model, and the *Bastankhah and PortéAgel* model. A

detailed comparison of wake models and their uncertainties can be found in Gao et al. (2020). The optimization models and solution methods presented in this section are independent of the wake model selected, since the wake effect is given as input coefficients.

We assume that the wake effect is *linearly additive*, meaning that the wake effect on a given turbine from a set of turbines is the sum of the individual contributions. Although some accuracy is lost (Renkema, 2007), since the sum of squares would be more correct, it is a reasonable approximation in practice.

Two versions of the WFLOP exist: in the *continuous* problem, all locations within a given area are available. In the *discrete* version, a candidate set of locations have been a-priori selected, and the task is to place the needed number of turbines at a subset of these positions. The candidate positions are typically forming a grid or mesh. Long et al. (2017) presented a comparison of the two models. The *continuous* problem is most often solved using iterative methods. An initial solution is generated, and the wake effect on each turbine is used to calculate a gradient minimizing the wake. Each turbine is then repeatedly moved a small step in the direction of the gradient. This approach generally works well if the available area $\mathcal{R}$ is convex, but if there are obstacles in the area, and different foundation costs, it can be challenging to make the algorithm converge.

The *discrete* version has the advantage that obstacles can be easily handled by removing forbidden positions from the set of available locations. Foundation costs can be handled by pre-calculating the cost of the foundation at the given water depth corresponding to the given location. Furthermore, the wake effect of a turbine located at position i on a turbine at location j can be pre-calculated for all pairs of locations. However, the discrete version of the problem is NP-hard since it contains the *p-dispersion sum problem* as a special case (Pisinger, 2006) by setting the minimum separation distance $d^{\min}$ to zero.

In the following we consider the *discrete* version, since it is independent of the chosen wake model, and does not require technical knowledge about fluid mechanics. Assume that we want to place T homogeneous turbines on a set of $\mathcal{N}$ available positions. For each position $i \in \mathcal{N}$ we introduce a binary variable x_i to denote whether a turbine is placed at the position. Let p_i be the expected energy production per year and f_i be the building costs (turbine and foundation), at location i . Furthermore, let w_{ij} be the expected lost energy production (due to wake) per year at location i if another turbine is located at position j . The loss can be calculated by considering all wind directions multiplied by the the average wind speed in the direction as given by the wind rose. If we pre-calculate w_{ij} for all pairs of locations i, j , then our objective becomes

$$\phi \left(\sum_{i \in \mathcal{N}} p_i x_i - \sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{N}} w_{ij} x_i x_j \right) - \sum_{i \in \mathcal{N}} f_i x_i, \quad (1)$$

where ϕ is the *Energy Price Factor* (EPF), which is used to convert the energy production to revenues (Cazzaro et al., 2023b). In calculating this factor, we consider the future yearly electricity energy prices, EP_t , for the lifetime T of the OWF, and a yearly interest rate r . The resulting energy price factor is thus calculated as: $\phi = \sum_{t=0}^{T-1} \frac{EP_t}{(1+r)^t}$. If an existing OWF is present near the planned farm, the wake effect of the existing turbines can be added to w_{ij} . If we, on the other hand, let

$$P_{ij} = \begin{cases} \phi p_i - f_i & \text{if } i = j \\ -\phi w_{ij} & \text{if } i \neq j \end{cases}, \quad (2)$$

then we reach the formulation (Cazzaro and Pisinger, 2022b):

$$\max \sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{N}} P_{ij} x_i x_j, \quad (3)$$

$$\text{s.t.} \quad \sum_{i \in \mathcal{N}} x_i = T, \quad (4)$$

$$x_i + x_j \leq 1, \quad i, j \in \mathcal{N}, \text{ dist}(i, j) \leq d^{\min}, \quad (5)$$

$$x_i \in \{0, 1\}, \quad i \in \mathcal{N}. \quad (6)$$

The quadratic objective function (3) maximizes the energy production revenues minus the building costs. Constraint (4) ensures that the desired number of turbines T is placed in the area, while constraints (5)

forbid placing turbines that are closer than the minimum distance $d^{\min}$ apart. Finally, we define the domain of the decision variables in constraints (6).

The main strength of the formulation is that all wake effect calculations can be precomputed in the P matrix. However, the objective function is quadratic, making it necessary to linearize the model for some solution methods. Furthermore, the wake effect is only approximately calculated.

Table 1 provides a schematic overview of the literature. It can be seen that solution methods for the discrete WFLOP roughly can be divided into the following groups:

MIP-models For smaller problems, the formulation (3)–(6) can be solved directly by commercial solvers. Since $P_{ii} \geq 0$, and $P_{ij} \leq 0$ for $i \neq j$, it is easy to linearize the objective function by introducing variables $y_{ij} \geq 0$ defined as $y_{ij} = x_i x_j$ (Donovan, 2005; Fagerfjäll, 2010). The objective function then becomes $\sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{N}} P_{ij} y_{ij}$, and we add the constraints $x_i + x_j - y_{ij} \leq 1$ for all $i, j \in \mathcal{N}$. Another linearization, proposed by Fischetti et al. (2016), is to introduce variables $w'_i \geq 0$ defined as $w'_i = (\sum_{j \in \mathcal{N}} w_{ij} x_j) x_i$. Then the objective function becomes $\sum_{i \in \mathcal{N}} (p_i x_i - w'_i)$, and we add the constraints $\sum_{j \in \mathcal{N}} w_{ij} x_j \leq w'_i + M(1 - x_i)$ for all $i \in \mathcal{N}$. Finally, Donovan (2005) noticed that by ignoring the wake effect w_{ij} , the objective function becomes a linear function $\max \sum_{i \in \mathcal{N}} p_i x_i$, which results in a vertex packing problem.

Matheuristics Fischetti and Monaci (2016) proposed a matheuristic based on *Proximity Search* to solve the formulation (3)–(6). The idea is to use a MIP-solver to find an initial solution by considering a simplified formulation without the interference constraints. Then, the objective function is modified such that it counts the number of variables changed from the current solution. An additional constraint is added, stating that the objective function should improve the current solution by a given threshold value Θ . The modified problem is then solved, and the process can be repeated.

Reformulation to dispersion problem The minimum distance constraint (5) can easily be handled by setting $P_{ij} = -M$ in equation (2) when two turbines are too close, i.e. $\text{dist}(i, j) \leq d^{\min}$. In this way, the problem becomes an ordinary *p-dispersion Sum Problem* (Donovan, 2005; Zhang et al., 2014). Greedy heuristics and bounds for the p-dispersion problem can be found in Pisinger (2006). Zhang et al. (2014) proposed some improvements to the dispersion model and showed that these made it possible to solve larger problems. Pisinger and Rustad (2025) included upper and lower bounds on cabling costs to P_{ii} , making it possible to guide the WFLOP towards cheaper cabling solutions.

Constraint programming Zhang et al. (2014) proposed using constraint programming (CP) to solve the WFLO, restricted to constraints (4) and (5). The main advantage of using CP was that it allowed the use of a more detailed (non-polynomial) wake model, since CP is based on enumeration and therefore accepts more general functions.

Metaheuristics A large number of metaheuristics have been proposed based on local search, evolutionary algorithms, and hyper-heuristics. Gualtieri (2020) gives an overview of heuristic solution methods: Genetic Algorithms (GA) (Mosetti et al., 1994), Particle Swarm Optimization (PSO) (Chowdhury et al., 2012), Evolutionary Algorithm (EA) (Mora et al., 2007), Viral System Algorithm (VSA) (Ituarte-Villarreal and Espiritu, 2011), or Extended Pattern Search (EPS) (DuPont et al., 2016). According to the author, GAs are used in more than 75% of WFLO studies. Thomas et al. (2023) compare eight heuristic optimization methods to a WFLOP (*Borssele III and IV*) with 81 OWTs.

In summary, huge progress has been made in solving the WFLO since the first work by Mosetti et al. (1994). We have seen a large variety of solution methods proposed during the last decades, each having their advantages and targeting specific constraints. MIP-models and matheuristics provide a solid foundation for solving the discrete version of the problem, but may experience long running times. On the other hand, advanced local search heuristics are able to handle more side constraints and can solve larger realistic problems, but cannot provide a solution guarantee. The recent work by Pisinger and Rustad (2025) may be a promising direction of research, as the algorithm combines flexibility from heuristics with tight upper bounds.

Paper	Strategy	Solution Method	Problem Instance
Mosetti et al. (1994)	Heuristic	GA	<i>Mosetti</i> 10x10 grid, 15-26 turbines
Ozturk and Norman (2004)	Heuristic	Greedy	46x46 grid
Grady et al. (2005)	Heuristic	GA	<i>Mosetti</i> 10x10 grid
Donovan (2005)	Exact	MIP (vertex packing / linearized)	500-2500 positions, 5-60 turbines
Huang (2007)	Heuristic	Distributed GA	<i>Mosetti</i> 10x10 grid
Mora et al. (2007)	Heuristic	GA	20x20 grid, 50 turbines
Elkinton et al. (2008)	Heuristic	Gradient Search, Greedy, GA, SA, PSA	20x20 grid, and <i>Hull, Massachussets</i> , 4 turbines
Huang (2009)	Heuristic	Hybrid GA	<i>Mosetti</i> 10x10 grid
Rivas et al. (2009)	Heuristic	SA	<i>HornsRev1</i> , 64-106 turbines
Fagerfjäll (2010)	Exact	MIP	<i>PARK</i> , 30x30 or 60x60 grid, 28-30 turbines
Şişbot et al. (2010)	Heuristic	Multi-objective GA	<i>Mosetti</i> 10x10 grid, and existing park
Emami and Noghreh (2010)	Heuristic	GA	<i>Mosetti</i> 10x10 grid
González et al. (2010)	Heuristic	GA	<i>Mosetti</i> 10x10 grid
Bilbao and Alba (2010)	Heuristic	CHC (GA), SA	10x10 grid, 30-50 turbines
Archer et al. (2011)	Exact	MILP	17x22 grid, 10-26 turbines
Yin and Wang (2012)	Heuristic	Grasp + VNS	<i>Mosetti</i> 10x10 grid
Chen et al. (2013)	Heuristic	Nested GA	20x20 grid, 41 turbines (5376 positions, 18 turbines)
Turner et al. (2014)	Exact	MILP, Quadratic Integer Programming	<i>Mosetti</i> 10x10 grid
Zhang et al. (2014)	Exact	MIP, Constraint Programming	20x20 grid, 40 turbines
Mittal et al. (2015)	Heuristic	Bi-level GA + Gradient based approach	57 positions, 8-11 turbines
Fischetti and Monaci (2016)	Matheuristic	Proximity search + MIP	1.000-20.000 positions, max profit (no limit turbines)
Parada et al. (2017)	Heuristic	GA	<i>Mosetti</i> , 20x20 grid
Biswas et al. (2017)	Heuristic	L-shade (Differential Evolution)	<i>Mosetti</i> 10x10 grid
Rezk et al. (2019)	Heuristic	Nature inspired optimization algorithms	<i>Mosetti</i> 10x10 grid
Patel et al. (2019)	Heuristic	Pattern-based approach	Up to 33 turbines
Fischetti and Pisinger (2019)	Matheuristic	Proximity search	<i>Borssele1</i> , above 10.000 positions, 50 turbines
Liu et al. (2020)	Heuristic	Bi-criteria GA	<i>HornsRev1</i> , 1989 positions
Cazzaro and Pisinger (2022b)	Heuristic	P-dispersion, VNS	<i>SyntheticA-J</i> 3248-21709 positions, 8-75 turbines, <i>Mosetti</i>
Cazzaro et al. (2023a)	Heuristic	P-dispersion, VNS	<i>SyntheticB</i> 6989 positions, 99 turbines, vertical/horizontal turbines

Table 1: Overview of selected papers for WFLO using the *discrete* formulation, ordered by publication year. *Strategy*: exact or heuristic. *Solution method*: Genetic Algorithm (GA), Simulated Annealing (SA), Particle Swarm Optimization (PSO), Ants Colony Optimization (ACO), Pattern Search Algorithm (PSA), Variable Neighborhood Search (VNS). *Problem instance*: WF, number of positions/turbines.

3.2. Wind farm cable routing

It is estimated that between 8% and 24% of the installed cost of an OWF is related to its electrical system (IRENA, 2024). The *wind farm cable routing problem* (WFCR) consists in finding the optimal network of inter-array cables that transfers the electricity produced by the wind turbines, subject to a number of constraints. The electricity is conveyed to one or more central *substation(s)*, which are connected onshore by a high-capacity export cable. If more than one substation is present, the problem also includes deciding which turbines should be connected to each substation. For ease of presentation we will only consider one substation in the following. In the WFCR we will not distinguish between offshore and onshore problems, since they are, in principle, the same.

Fagerfjäll (2010) presented the first MIP-model for the WFCR with multiple cable types. Bauer and Lysgaard (2015) introduced no-crossing constraints for cables, while Pillai et al. (2015) introduced ways of handling obstacles in the area. Cerveira et al. (2016) were the first to take power losses into account, while Cazzaro and Pisinger (2022a) introduced balancing constraints on the cables entering the substation. See Hou et al. (2019) for a recent survey, and Cazzaro and Pisinger (2022a) for a detailed bibliography.

It is assumed that the position of the turbines and the substation are fixed and given as input to the problem. Furthermore, a set of *cable types* is given as input. Each cable type has a cost per meter, and a given capacity. For ease of presentation we assume that only one substation is present, although we will discuss generalizations at the end of this section.

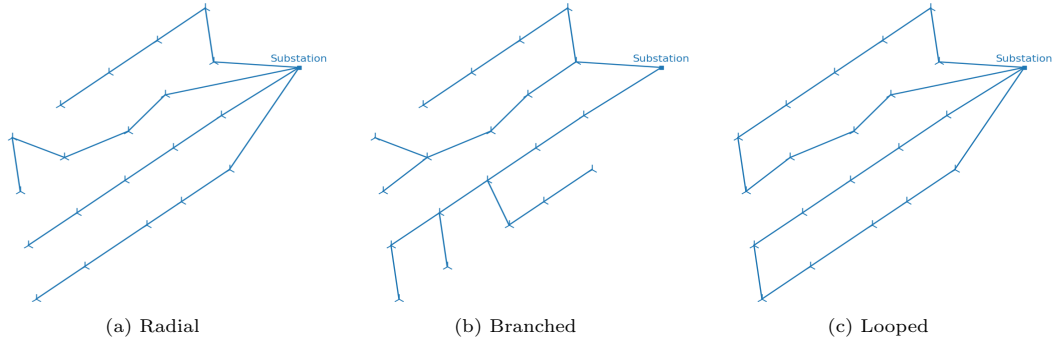


Figure 4: Examples showing three different topologies for cable routing, for same turbine positions (Cazzaro and Pisinger, 2022a)

The cabling is subject to a given *topology* (see Figure 4): In the *branched* version, a turbine can have multiple entering cables, and exactly one exit cable; thus, turbines are connected in a tree. In the *radial* version (also called *string* or *spoke graph* version), all turbines have at most one entering cable and exactly one exit cable; thus, turbines are connected in strings. In the *looped* version, all cables are organized in loops starting and ending at the substation. Although the branched version will lead to the lowest cabling costs, having multiple cables entering a turbine imposes additional costs. We will call it *penalty branched* topology, if a penalty is paid for each additional cable. To avoid extra costs, the radial version is often preferred in practice. The looped version has the advantage that if a cable breaks down, every turbine is still connected to the network. This is important, since the turbine needs power to perform a safe shut-down. Fischetti and Pisinger (2018a) presents a detailed study of how the various topologies can be formulated in a MIP model and how they affect the obtained solution.

Obstacles are present in nearly all WFs, increasing the complexity of finding the optimal configurations for the inter-array cables. Obstacles can be caused by soil conditions, shipwrecks, existing cables, sailing corridors, or environmental protection. Obstacles can be both concave and convex. Cables are not allowed to pass inside obstacles. In most cases, we also have an *outer boundary* that cables are not allowed to cross. Obstacles in the available plan are typically handled by introducing *Steiner points* around the corners of the forbidden area (Fischetti and Pisinger, 2018a). The cabling problem is allowed to use Steiner points, but they do not need to be visited. Next, all edges that cross forbidden areas are removed in a pre-processing step. An alternative method to handle obstacles is to generate a *visibility graph* (Welzl, 1985), and then calculate all distances using the visibility graph, as proposed by Cazzaro and Pisinger

(2022a). In most models, it is assumed that cables do not *cross* each other, since this may generate excess heat that can damage cables. Furthermore, cable crossings make maintenance more complicated.

A given set $\mathcal{K}$ of *cable types* is available, where c^k is the cost per meter of a cable type supporting up to k turbines, $k \in \mathcal{K}$. Additional “virtual” cable types can be calculated as combinations of existing cables for all $k = 1, \dots, N$ using a dynamic programming recursion for the knapsack problem (Kellerer et al., 2004). A common constraint is to limit the number of *connections to the substation* by C . This constraint is motivated by the fact that cables need to have some separation distance. Another common constraint is to *balance* the load on each cable entering the substation. Typically, the number of turbines connected to each cable entering the base may differ by at most one turbine. This constraint makes it possible to use standardized transformers on the substation. Finally, we have *flow conservation constraints*, ensuring that electricity produced by the turbines is transferred to the substations. In addition, the cable type used in each connection must have enough *capacity* to transfer the electricity produced by its turbine plus all upstream turbines in its string.

In order to model the WFCR formally, let $\mathcal{T}$ be the set of turbines to connect and $\mathcal{T}_0 := \mathcal{T} \cup \{0\}$, where position 0 is the substation. The length d_{ij} of a cable connecting turbines i and j is computed as the Euclidean distance between the turbines. The cable cost is c^k per meter of a cable supporting k turbines.

In the following model, we assume a *branched* topology. Other topologies are discussed after the model. The decision variables y_{ij}^k indicate whether a turbines i and j are connected using cable k , while the continuous variables f_{ij} denote the power flow (measured in number of turbines) on edge (i, j) . A basic formulation by Cazzaro and Pisinger (2022a) is then

$$\min \sum_{i \in \mathcal{T}_0} \sum_{j \in \mathcal{T}_0} \sum_{k \in \mathcal{K}} d_{ij} c^k y_{ij}^k, \quad (7)$$

$$s.t. \sum_{k \in \mathcal{K}} y_{ij}^k \leq 1, \quad i, j \in \mathcal{T}_0, \quad (8)$$

$$\sum_{j \in \mathcal{T}_0} \sum_{k \in \mathcal{K}} y_{ij}^k = 1, \quad i \in \mathcal{T}, \quad (9)$$

$$\sum_{j \in \mathcal{T}_0} \sum_{k \in \mathcal{K}} y_{0j}^k = 0, \quad (10)$$

$$y_{ii}^k = 0, \quad i \in \mathcal{T}_0, k \in \mathcal{K}, \quad (11)$$

$$f_{ij} \leq \sum_{k \in \mathcal{K}} k y_{ij}^k, \quad i, j \in \mathcal{T}_0, \quad (12)$$

$$\sum_{j \in \mathcal{T}_0} f_{ij} = \sum_{l \in \mathcal{T}_0} f_{li} + 1, \quad i \in \mathcal{T}, \quad (13)$$

$$\sum_{i \in \mathcal{T}} f_{i0} = |\mathcal{T}|, \quad (14)$$

$$\sum_{i \in \mathcal{T}} \sum_{k \in \mathcal{K}} y_{i0}^k = W, \quad (15)$$

$$\sum_{k \in \mathcal{K}} y_{ij}^k + \sum_{k \in \mathcal{K}} y_{i'j'}^k \leq 1, \quad [(i, j), (i', j')] \in \mathcal{C}, \quad (16)$$

$$y_{ij}^k \in \{0, 1\}, \quad i, j \in \mathcal{T}_0, k \in \mathcal{K}, \quad (17)$$

$$f_{ij} \geq 0, \quad i, j \in \mathcal{T}_0. \quad (18)$$

The objective function (7) minimizes the cost of all cables used. The first constraints (8) make sure that only one cable type can be selected for each connection. The second constraints (9) ensure that each turbine has one exit cable, except for the substation 0, which has none (10), since we do not consider the export cable in the formulation. Constraints (11) forbid cable loops that connect a turbine to itself. Constraints (12) make sure that the power flow in each segment does not exceed the cable capacity. Constraints (13) track the power flow in the cables, ensuring that each flow variable reports the correct

power in each cable. Constraint (14) guarantee that all the flow reaches the substation. We ensure with constraint (15) that W number of cables is connected directly to the substation. Constraints (16) forbid cable crossings in the routing solution, in which $\mathcal{C}$ is the set of all cable pairs that cross each other. Lastly, constraints (17) and (18) specify the domain of variables.

The main advantage of the above model is the fact that it can easily be modified to handle various topologies. However, the 3-index formulation leads to many variables, and the non-crossing constraints grow quadratically with the number of positions.

Fischetti and Pisinger (2018b) prove the NP-hardness of the problem even in some relevant special cases. Other topologies can easily be incorporated in the above model as shown in Fischetti and Pisinger (2018a): To model the *radial* topology we add the following constraints

$$\sum_{j \in \mathcal{T}} \sum_{k \in \mathcal{K}} y_{ji}^k \leq 1, \quad i \in \mathcal{T}, \quad (19)$$

ensuring that at most one cable can enter each turbine i . Since constraints (9) ensure that one cable leaves each turbine, cables will form strings.

To model the *looped* topology we introduce extra binary variables $q_{ij} \in \{0, 1\}$ if a redundant cable is installed between turbines i and j . We add the following constraints

$$\sum_{j \in \mathcal{T}} \left(\sum_{k \in \mathcal{K}} y_{ji}^k + q_{ji} \right) = 1, \quad i \in \mathcal{T}, \quad (20)$$

to ensure that exactly one cable (either a real cable or the redundant cable) enters turbine i . We also need to pay for the redundant cables in the objective function.

$$\min \quad \sum_{i \in \mathcal{T}_0} \sum_{j \in \mathcal{T}_0} \sum_{k \in \mathcal{K}} d_{ij} c^k y_{ij}^k + \sum_{i \in \mathcal{T}} \sum_{j \in \mathcal{T}} d_{ij} c^{k'} q_{ij}, \quad (21)$$

where k' is the cheapest cable type, to be used for the redundant cable.

Cerveira et al. (2016) were the first to consider *power losses* in the formulation, considering both voltage drop and line losses. Fischetti and Pisinger (2018b) observed that if the turbines have a similar production over a year, then the power losses can be handled by preprocessing. For each cable of type t supporting exactly k turbines we can introduce a new virtual cable type $t' \in \mathcal{T}'$, whose cost is equal to the cost of cable type t minus the cost of power losses over the lifetime of the OWF of k turbines. Then the MIP formulation (7)–(18) can be solved using the full set of virtual cable types $\mathcal{T}'$.

Solution methods for the cabling problem can roughly be divided into the following categories. Some of the approaches only work for a specific topology, in which case we indicate the topology in parentheses (see Table 2 for a literature overview).

MIP-models (all topologies) The MIP formulation (7)–(18), with corresponding topology extensions, can be solved to optimality for small to moderately sized problems by commercial solvers. Since the non-crossing constraints contain $O(n^4)$ constraints, most authors solve the MIP formulation without the non-crossing constraints, and only add these constraints if they are violated. This can be seen as a cutting-plane algorithm (Fischetti and Monaci, 2016). Donovan (2005) used an alternative formulation using vertex packing, and solved the problem to optimality.

Matheuristics (all topologies) Fischetti and Pisinger (2018a,b) proposed a hybrid matheuristic, in which a matheuristic is used several times, and then the best found solution is passed to a MIP solver to solve the problem to optimality (with a time limit). The matheuristic phase works by temporarily fixing some selected y_{ij} variables to 1 and removing all decision variables that will result in a cable crossing with the selected cables. Then, a MILP solver is used to solve the corresponding restricted problem using a short time limit.

Minimum spanning tree (MST) (branched topology) An obvious heuristic approach is to solve a minimum spanning tree as, e.g., in Cazzaro et al. (2020). Next, a depth-first search from the root (substation) is used to calculate the number of turbines below each node. Cable types are then selected to accommodate the needed power. However, an MST will frequently have very few connections out from the substation, making it difficult to find sufficiently large cables. Also, it is worth noticing that a MST with the cheapest cable provides a lower bound on the overall cabling costs.

Paper	topology	non-cross.	pow.loss	obstacles	substation	Problems solved	Solution method
Fagerfjäll (2010)	branched	no	no	no	M	<i>PARK</i> , 28-30 turbines, 2 cable types	MIP-model
Bauer and Lysgaard (2015)	radial	yes	no	no	M	<i>Barrow</i> , <i>Walney1</i> , <i>Sheringham</i> , 30-88 turbines, 2 cable types	Open VRP formulation, MIP model, Clark-wright savings heuristic
Pillai et al. (2015)	branched	yes	no	yes	ML	<i>NavitusBay</i> , 175 turbines, 1 cable type	MIP model, Delaunay triangulation
Cerveira et al. (2016)	branched	no	yes	no	S	3 Portuguese WFs, 25-57 turbines, 12 cable types	MIP-model, branch-and-cut
Hou et al. (2017b)	branched	yes	yes	no	M	<i>Anholt</i> , 111 turbines, 5 cable types	adaptive PSO-MST algorithm
Fischetti and Pisinger (2018b)	branched	yes	yes	yes	ML	<i>HornsRev1</i> , <i>KetishFlats</i> , <i>Ormonde</i> , <i>DanTyisk</i> , <i>Thanet</i> , 30-100 turbines, 2-5 cable types	MIP, hybrid matheuristic
Fischetti and Pisinger (2018a)	radial, branched, looped	yes	no	yes	ML	5 real WF from DK+UK, 50-100 turbines, 2-3 cable types	MIP, hybrid matheuristic
Jin et al. (2019)	branched	yes	yes	no	S	<i>Anholt</i> 111 turbines, 4 cable types, 1 substation	APSO-SST
Qi et al. (2019)	radial	yes	yes	no	S	50 turbines, 11 cable types	open multiple TSP formulation, Modified Bat Algorithm
Cazzaro et al. (2020)	branched	yes	no	no	SL	5 real-life farms, 50-120 turbines, 3 cable types	SA, TS, GA, VNS, MST, sweep-line
Cazzaro and Pisinger (2022a)	radial, branched	yes	yes	yes	MLB	10 synthetic, 14-119 turbines, 2 cable types	visibility graph, LNS (swap, double-swap, cycle-swap, re-partition)
Taylor et al. (2022)	penalty branched	penalty/constraint	yes	Steiner nodes	ML	synthetic instance, 10-122 turbines, 3 cable types	ant colony heuristic

Table 2: Overview of selected papers for the inter-array cabling, ordered by publication year. **Topology** (Radial, Branched, Looped) Penalty for branches, limit on branches; **Crossing** allowed (Yes, No, Penalty); **Power loss** (Yes, No) pre-calculated; **Obstacles** (No, Steiner points, visibility graph); **Substation** S (Single substation), M (Multiple substations), L (Limited cables to substation) B (Balanced power load); **Problem instance** Number of turbines, cable types, substations; **Solution method** (MIP, matheuristic, sweep-line, MST (minimum spanning tree), VNS (variable neighborhood search), APSO (adaptive particle swarm optimization), SST (stochastic spanning tree)).

Capacitated MST (branched topology) Cerveira et al. (2016) proposed a MIP formulation inspired by the capacitated MST (Gouveia, 1995). To strengthen the formulation, a number of valid inequalities was added. This included the inequalities from Gouveia (1995), as well as new inequalities based on bounding the number of downstream connections with a given capacity (Cerveira et al., 2016).

Cost scaling (all topologies) Cazzaro and Pisinger (2022a) proposed using a cost-scaling method based on the framework by Katayama (2019) for the capacitated multi-commodity flow problem. The idea is to solve the MIP-model (7)–(18) while relaxing the binary constraints (17) to $0 \leq y_{ij}^k \leq 1$. This makes the problem easier to solve, but the resulting LP-solution will often be fractional. Typically, a small fraction of the thickest cable is used, since it has the lowest cost per power unit. The idea in the cost scaling method is to adjust costs for the used cables, such that the current LP-solution will end up paying for the full cable. After adjusting the costs, the relaxed problem is solved again, and the process is repeated until a sufficiently good solution has been found.

VRP-inspired solution methods (radial topology) Bauer and Lysgaard (2015) used an Open VRP formulation to solve the problem. A hop-indexed MIP model is proposed, and it is solved using a branch-and-cut framework, where planarity constraints are added only when they are violated. To speed up the solution process, a good initial solution is constructed using the Clarke and Wright savings heuristic, followed by a 2-opt exchange heuristic.

Sweep line methods (all topologies) Cazzaro et al. (2020) and Cazzaro and Pisinger (2022a) proposed a rotating sweep line heuristic to initially split the turbines into subtrees. With the substation as the center, a rotating sweep line is used to partition the turbines into sets of equivalent size. Next, each subset is connected to the substation by solving a MIP model or using heuristics. Depending on the start position of the rotating sweep line, one can obtain different solutions, hence a multistart approach may lead to improved solutions.

Metaheuristics (all topologies) Several metaheuristics have been proposed. Local search methods typically remove and insert connections between turbines. Cazzaro et al. (2020) compared five different heuristics based on Simulated Annealing (SA), Tabu Search (TS), Variable Neighborhood Search (VNS), multistart sweep line, and Genetic Algorithm (GA). In the computational experiments, VNS performed best, followed by TS.

Overall, we see great progress in solution methods and in how expressive the models have become. Several real-life constraints can now be handled in the models, making the solution methods applicable in practice. Some of the recent papers, such as Cazzaro and Pisinger (2022a), are able to solve large-sized instances with more than 100 turbines in less than 3-4 hours. Smaller instances are solved within a few minutes.

Metaheuristics generally scale better than MIP-models or Matheuristics, making them more applicable for large-scale problems. Heuristics are also generally better at handling real-life constraints that can be difficult to formulate in an MIP-model. Among the above solution methods, it will often be the considered topology that determines which method is most suitable, as it is essential to utilize structural properties.

We have so far assumed that only one substation is present. If *multiple substations* are present, the problem also includes an assignment problem to decide which turbines to connect to which station. Hou et al. (2017b) proposed using a fuzzy C-means clustering method to assign turbines to substations. Fagerfjäll (2010) handled multiple substations as part of the proposed MIP model for cable routing.

3.3. Integrated wind farm layout and cable routing

The WFLO and the WFCR have contradicting objectives: the WFLO aims at spreading out the turbines as far from each other as possible, in order to reduce the wake effect. On the other hand the WFCR favors solutions that have turbines close to each other to minimize connection costs. Therefore, a sequential approach, where first the WFLO is solved and then the WFCR is optimized, may lead to non-optimal, or even infeasible, solutions.

Therefore, solving the two problems as an integrated process is an obvious approach to reach lower cost of produced energy. However, since both the WFLO and the WFCR are difficult NP-hard problems, an integrated approach may be very difficult to solve, meaning that only tiny problems can be solved to optimality. Using heuristics may scale better, but exploring the much larger solution space can be difficult, and can, in practice, result in worse solutions than a sequential approach.

Paper	Integration	Overall framework	Solution method WFLO	Solution method WFCR	Problems solved
Fagerfjäll (2010)	unclear	MIP-model	MIP-model	MIP-model	none reported
Wu et al. (2013)	master-slave	genetic algorithm	genetic algorithm	Ant Colony	8x8 grid, 32 turbines
Hou et al. (2017c)	master-slave	Particle swarm	continuous	variant of MST	<i>Norcowe</i> , <i>HornsRev1</i> instances, 80 turbines, 6 cable types
Tao et al. (2020a)	master-slave	bi-level multi-objective	genetic algorithm	binary particle swarm / quadratic programming	<i>GreaterGabbard</i> instance, 12x12 positions, 73-80 turbines, 5 cable types
Wu et al. (2020)	master-slave	genetic algorithm	genetic algorithm	GeoSteiner	10x10 grid, 2-58 turbines
Fischetti and Fischetti (2023)	full	MIP-model, branch and cut	MIP-model	MIP-model	synthetic instances, 40-100 positions, 10-30 turbines.
Cazzaro et al. (2023b)	full	MIP model, VNS local search	1-opt, 2-opt, shake	swap, matheuristic	10 synthetic instances, 3.238-21.709 positions, 20-235 turbines
Butterwick et al. (2023)	full	Hyper-heuristic	10 low-level heuristics	10 low-level heuristics	4 <i>Borssele</i> instances, 55-283 positions, 5-40 turbines, 7 cable types
Pisinger and Rustad (2025)	semi-integrated	Expected cable costs included in WFLOP; bounds for full problem	VNS local search	matheuristic	10 synthetic instances, 3.238-21.709 positions, 20-235 turbines

Table 3: Overview of selected papers for integrated WFLO and WFCR problems, ordered by publication year. **Integration** Master-slave, fully coupled, semi-integrated **Solution method** MIP-model, matheuristic, MST (minimum spanning tree), VNS (variable neighborhoods search).

Fagerfjäll (2010) was the first to propose a combined approach for the WFLO and WFCR. However, the conclusion was that the combined problem does seem to be very complex, and the problems possible to solve within a reasonable amount of time were too small to be of interest. With improved solution methods, the combined approach has recently become more attractive, and a number of papers have been presented in the recent years.

Table 3 gives an overview of papers studying the combined WFLO and WFCR problem. *Master-slave* means that the current solution only contains information about the WFLO. In every iteration, the WFCR is solved from scratch to return an overall objective. The solution methods can briefly be classified as follows:

Exact methods Fischetti and Fischetti (2023) presented an integrated MIP model, where cuts are used to tighten the formulation. The overall model is solved through branch-and-cut. A number of projected cuts are presented, including quadratic cuts, clique cuts, and enhanced clique cuts. Pisinger and Rustad (2025) presented a number of efficient upper and lower bounds for the integrated problem and demonstrated that the gap on average is around 5% for realistic problems.

Local search heuristics Cazzaro et al. (2023b) proposed a combined VNS local search, where the solutions of the WFLO and the WFCR are modified simultaneously. Pisinger and Rustad (2025) presented a semi-integrated approach where upper and lower bounds on the expected cabling costs to each available turbine location are used to improve the WFLOP formulation (3)–(6). In this way the WFLOP can use predicted cabling costs to avoid expensive locations. The WFLOP was solved using a VNS heuristic, and the following WFCRP was solved using a matheuristic. Butterwick et al. (2023) proposed a hyper-heuristic, where 10 low-level heuristics are used to modify the WF layout and/or cabling. The neighborhoods consist of inserting/moving both turbines and cables, where some greedy strategies are used to connect the cables (e.g., connect a new turbine to the nearest turbine). Several selection methods are studied for choosing a low-level heuristic. Tao et al. (2020a) formulated the problem as a bi-level multi-objective optimization, and proposed a genetic algorithm for WFLO (non-dominated sorting genetic algorithm) and two heuristics for WFCR (binary particle swarm optimization and quadratic programming).

Overall, the metaheuristics by Cazzaro et al. (2023b) and Pisinger and Rustad (2025) are best at solving large problems and has the flexibility to model many real-life constraints. The last algorithm is particularly interesting because it also returns tight bounds on the solution.

3.4. Handling uncertainty in offshore wind farm design

The design of OWFs is strongly affected by uncertainties in wind and weather. Furthermore, energy prices may fluctuate, adding further uncertainty to the decisions. Since wind turbines and cables cannot be moved once they are installed, no recourse options are available, and the stochastic problems therefore become single-stage problems. Many of the uncertainties can be incorporated in the models through preprocessing, without significantly changing the formulation.

For instance, the wake effect w_{ij} in equation (1) can be calculated as the expected value over all wind scenarios. Fischetti and Monaci (2016) consider 500 macro-scenarios computed on real-world wind data from Vattenfall. In a similar way, different energy price scenarios can be considered when calculating the expected earning of a turbine.

Wind scenarios not only affect the energy production of a given turbine, but also implicitly the cable losses from resistance due to varying electric current. Agra and Cerveira (2024) consider the cabling problem under uncertain wind scenarios, while Fischetti and Pisinger (2018b) pre-computed the cable losses as averages over all wind scenarios, resulting in a standard, deterministic, WFCRP.

3.5. Data instances

Despite the large amount of research done in this field, there are only a few standard data sets in the literature. Mosetti et al. (1994) presented a number of OWF layout instances on a 10×10 grid. By current standards, these instances are very small. Cazzaro et al. (2022) presented ten large-scale OWF layout instances, denoted instance A–J, having 3238 to 21709 positions. The instances contain obstacles and several zones. These instances were supplemented by Pisinger and Rustad (2025) who provided a wake matrix and cable information, so that the full data is available for solving WFLOP and WFCRP. In addition to these standardized data sets, many papers use data from real OWFs as a basis for their test instances, with Anholt, Borssele, HornsRev1, and PARK being used in multiple studies.

4. Installation

Once the layout of the OWF is decided, the next phase of the life cycle is to install all the components that make up the OWF. Planning the installation of an OWF includes setting up a supply chain to transport all the components from their producers to the staging port, selecting and designing the layout of such a port, and planning the logistics from the port to the wind farm. Optimization models and methods related to supply chain optimization and port layout are, to the authors knowledge, only addressed by Irawan et al. (2018) and Irawan et al. (2017c), respectively, and are not considered further in this article. The focus of this section is thus on the logistical planning of the last leg, from the staging port to the OWF itself, where a number of OWTs must be installed and connected by cables. In addition, one or more offshore substation(s) must also be installed, however, this is usually done prior to installing the OWTs themselves (Amorosi et al., 2024) and may therefore be considered separately.

The first component to be installed for an OWT is the foundation. Once the foundation is in place, installation of the array cables can begin. The installation of these cables involves three main activities: cable laying, cable burial, and cable pull-in, where the latter refers to the connection of the cable to either a turbine, substation, or shore (BVG Associates (2019)). It is common to install the array cables during foundation installation, while installation of export cables can be considered as an independent operation.

The third and final component to be installed is the turbine itself. A turbine consists of four main sub-components: tower, nacelle, hub, and blades, that are lifted from the deck of an installation vessel, or an adjacent transport vessel, to the OWT and connected (see Figure 2a). There exist three main strategies for installing the turbine. The first is the *individual parts* strategy, where each part of the turbine is installed at the OWF, without any pre-assembly in port, resulting in six lifting operations (tower, nacelle, hub and three blades). The second and third strategies for installing the turbine are known as *bunny ear* and *full rotor star*, respectively, where the former consists of pre-assembling two blades with the nacelle giving four lifts, and the latter consists of pre-assembling all three blades, and the hub, to the nacelle leading to only two lifting operations (Vis and Ursavas, 2016; BVG Associates, 2019).

An overview of the relevant literature presenting mathematical models for the installation phase of an OWF is given in Table 4. In the following, we first describe a mathematical model for scheduling a single installation vessel tasked with installing an OWF in Section 4.1. We then discuss extensions to several installation vessels, often in combination with fleet size and mix decisions, in Section 4.2, before giving an overview of models that consider installation planning as a project scheduling problem, with less focus on vessel operations, in Section 4.3. In Section 4.4 we discuss how uncertainty is incorporated in the different models, before discussing the solution methods used to solve different problems in the installation phase in Section 4.5. Finally, we give an overview of the data instances used in Section 4.6.

4.1. Scheduling of a single installation vessel

Traditionally, the same vessels were used for foundation and turbine installation. Scholz-Reiter et al. (2010) present the first MIP-model for a problem in the installation phase of the OWF life cycle. The problem considers a single vessel tasked with installing an OWF. They consider the installation of two components: foundations and turbines (referred to in their paper as *sub-structures* and *top-structures*). The objective is to create a schedule for the installation of the OWF that minimizes the total time it takes until the OWF is completed. The model is inspired by multi-period production planning and single machine job shop scheduling models, where the vessel is the machine processing jobs (installing components) and the production consists of installing components. Ursavas (2017) extends this model to a two-stage stochastic programming model, where uncertainty in weather conditions is taken into account.

Most recently, Bae and Ko (2023) present a simulation model for the installation of an OWF, where a MIP model, similar to that of Scholz-Reiter et al. (2010), is used to solve the planning problem of a single round trip for one vessel, tasked with installing a pre-defined set of components under given weather conditions. The paper distinguishes itself from the previous two by considering the installation of three components: foundation, tower, and top-structure (which in this case is the turbine except for the tower), and by explicitly modeling each turbine.

In this paper, we present a generalized version of the model proposed by Scholz-Reiter et al. (2010) that handles any number of components, loading sets, and number of installation round trips. Assume

that we have a set of C turbines that are to be installed at an OWF. For each turbine, we have a set of components $\mathcal{K}$, and we assume that each component $k \in \mathcal{K}$ takes B_k units of time to install. We further assume that the set $\mathcal{K} = \{1, 2, \dots, |\mathcal{K}|\}$ is ordered according to the sequence in which the components must be installed, so that component k must be installed before $k + 1$.

It is assumed that the installation of the components is done by a single vessel, performing multiple round trips between an onshore port and the OWF. Let $\mathcal{I} = \{1, \dots, n\}$ be the set of potential round trips. Before starting a round trip, the vessel is loaded with components, which it transports to the OWF and then installs. Let T be the travel time between the port and the wind farm, and let $\mathcal{L}$ be the set of loading sets, where each loading set contains a combination of components that can fit onto the vessel. We let L_{kl} denote the number of component $k \in \mathcal{K}$ in loading set $l \in \mathcal{L}$. Each round trip $i \in \mathcal{I}$ consists of a sequence of component installations, henceforth referred to as *installation activities*, $\mathcal{J} = \{1, \dots, m\}$, where $m = \max_{l \in \mathcal{L}} \{\sum_{k \in \mathcal{K}} L_{kl}\}$, i.e. the maximal number of components that can be installed on a single round trip.

Weather plays a critical role when installing OWT components. Due to safety considerations and vessel capabilities, each component and vessel combination has a threshold with respect to wave height and wind speed that limits when the installation may be performed. Let the planning horizon be divided into a set of weather periods $\mathcal{P}$, where W_p^S and W_p^E denote the start- and end-time of a weather period $p \in \mathcal{P}$. These periods are defined as time intervals in which the same set of components can be installed. Let $\mathcal{K}_p$ be the set of components that can be installed in weather period p , and let $\bar{\mathcal{K}}_p$ be the complement of this set, i.e., the set of components that may not be installed in weather period p .

Let z_i be a binary variable equal to 1 if round trip i is used, and 0 otherwise, while y_{il} is a variable equal to 1 if loading set l is chosen on round trip i , and 0 otherwise. The variable x_{ijk} is equal to 1 if component k is installed as installation activity number j on round trip i , and 0 otherwise. Furthermore, the variables t_i^S and t_i^E denote the start and end-times of round trip i , while the variable t_{ij} denotes the start time of installation activity j on round trip i . In addition, let the variable f_{ijk} denote the number of turbines, where the next component to be installed is component k , before starting installation activity j on round trip i . This variable is used to ensure that the components are installed in the correct order. Finally, let the variable w_{ijp} be equal to 0 if installation activity j on round trip i is completed before, and 1 if it starts after, weather period p . Note that if the component installed in activity j can be installed during this weather period, the corresponding variable w_{ijp} takes the value 0. Using this notation, the model may be formulated as follows:

$$\min t_n^E + \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} t_{ij}, \quad (22)$$

$$\sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} x_{ijk} = C, \quad k \in \mathcal{K}, \quad (23)$$

$$\sum_{k \in \mathcal{K}} x_{ijk} \leq z_i, \quad i \in \mathcal{I}, j \in \mathcal{J}, \quad (24)$$

$$z_{i+1} \leq z_i, \quad i \in \mathcal{I} \setminus \{n\}, \quad (25)$$

$$\sum_{j \in \mathcal{J}} x_{ijk} = \sum_{l \in \mathcal{L}} L_{kl} y_{il}, \quad i \in \mathcal{I}, k \in \mathcal{K}, \quad (26)$$

$$\sum_{l \in \mathcal{L}} y_{il} = 1, \quad i \in \mathcal{I}, \quad (27)$$

$$f_{i(j+1)k} = f_{ijk} + x_{ij(k-1)} - x_{ijk}, \quad i \in \mathcal{I}, j \in \mathcal{J} \setminus \{m\}, k \in \mathcal{K} \setminus \{1\}, \quad (28)$$

$$f_{(i+1)1k} = f_{imk} + x_{im(k-1)} - x_{imk}, \quad i \in \mathcal{I}, k \in \mathcal{K} \setminus \{1\}, \quad (29)$$

$$f_{11k} = \begin{cases} 1 & \text{if } k = 1 \\ 0 & \text{otherwise} \end{cases}, \quad k \in \mathcal{K}, \quad (30)$$

$$t_i^E + T z_{i+1} \leq t_{i+1}^S, \quad i \in \mathcal{I} \setminus \{n\}, \quad (31)$$

$$t_i^S + T z_i \leq t_{i1}, \quad i \in \mathcal{I}, \quad (32)$$

$$t_{ij} + \sum_{k \in \mathcal{K}} B_k x_{ijk} \leq t_{i(j+1)}, \quad i \in \mathcal{I}, j \in \mathcal{J} \setminus \{m\}, \quad (33)$$

$$t_{im} + \sum_{k \in \mathcal{K}} B_k x_{imk} \leq t_i^E, \quad i \in \mathcal{I}, \quad (34)$$

$$t_{ij} + \sum_{k \in \mathcal{K}_p} B_k x_{ijk} \leq W_p^S + M(w_{ijp} + \sum_{k \in \mathcal{K}_p} x_{ijk}), \quad i \in \mathcal{I}, j \in \mathcal{J}, p \in \mathcal{P}, \quad (35)$$

$$t_{ij} \geq W_p^E - M(2 - w_{ijp} - z_i), \quad i \in \mathcal{I}, j \in \mathcal{J}, p \in \mathcal{P}, \quad (36)$$

$$z_i \in \{0, 1\}, \quad i \in \mathcal{I}, \quad (37)$$

$$y_{il} \in \{0, 1\}, \quad i \in \mathcal{I}, l \in L, \quad (38)$$

$$t_{ij} \geq 0, \quad i \in \mathcal{I}, j \in \mathcal{J}, \quad (39)$$

$$f_{ijk} \geq 0, \quad i \in \mathcal{I}, j \in \mathcal{J}, k \in \mathcal{K}, \quad (40)$$

$$x_{ijk} \in \{0, 1\}, \quad i \in \mathcal{I}, j \in \mathcal{J}, k \in \mathcal{K}, \quad (41)$$

$$w_{ijp} \in \{0, 1\}, \quad i \in \mathcal{I}, j \in \mathcal{J}, p \in \mathcal{P}. \quad (42)$$

The objective function (22) minimizes the return time of the last round trip, which is also the completion time of the wind farm, and the sum of all start times for the installation of the components. This is the objective function used by Scholz-Reiter et al. (2010) and similar to that of Ursavas (2017), who also sums the start times of all round trips performed by the vessel. Even though the value provided by such an objective function does not make sense intuitively, it has two advantages: a) it removes symmetry from the model, and b) it gives the model an incentive to finish activities as early as possible, also when it does not improve the total installation time of the OWF. Bae and Ko (2023), on the other hand, who only consider one round trip, minimize only t_n^E .

Constraints (23) ensure that the correct number of each component is installed at the wind farm, while constraints (24) state that each installation activity can only be performed once and only if the round trip is used. Furthermore, constraints (25) make sure that a round trip is used only if the preceding round trip has been used. Constraints (26) limit the number of components installed on each round trip not to exceed the number of the corresponding component in the loading set chosen on that round trip, while constraints (27) force the model to choose exactly one loading set per round trip. Furthermore, constraints (28)–(30) are bookkeeping constraints keeping track of how many of each component may be installed as installation activity j on round trip i . Note that the first component installed on the first round trip must be a component of type 1.

Constraints (31)–(36) handle the update of time during the planning horizon. Constraints (31) state that if round trip $i + 1$ is used, then the start time has to exceed the end time of round trip i and the time the vessel uses to return to port, while constraints (32) ensure that the earliest time the vessel can start installing components on round trip i is the start time of the round trip and the travel time back to the wind farm. Constraints (33) ensure that installation activity $j + 1$ does not start before installation activity j is completed, and constraints (34) make sure that the vessel does not return until after the last installation activity. Furthermore, constraints (35) and (36) ensure that no installation activities are performed during time periods when the weather conditions are not suitable for the particular activity. In addition, we would like to point out that if not all n round trips are used, the model will force $t_n^E = t_{nm} = \dots = t_{n1} = t_n^S = t_{n-1}^E$ and so on. Finally, constraints (37)–(42) provide the variable domains.

The main advantage of this model is that it is based on a model for the well-known model job shop scheduling problem, for which there has been proposed numerous effective solution methods. Furthermore, the complexity of the model grows linearly when increasing the number of turbines or components. The disadvantages are that the LP-relaxation is quite weak, that it does not easily extend to multiple vessels (see next section), and that it does not consider the positions of the individual wind turbines.

4.2. Scheduling of multiple installation vessels

Due to increasing component sizes and weights, specialized vessels have become available for different installation activities in recent years (BVG Associates (2019)). Thus, many papers have looked at the problem where a set of vessels $\mathcal{V}$ has to be coordinated to install an OWF, with varying assumptions about capabilities of each vessel. Backe and Haugland (2017), Rippel et al. (2019) and Amorosi et al. (2024)

consider models in which each vessel only performs one installation activity. Rippel et al. (2019) consider installing the turbine as the only installation task, while Backe and Haugland (2017) let different vessels represent different installation strategies for a turbine, i.e., individual parts, bunny ear, and *feed*, where the latter consists of a vessel staying at the OWF at all times, while barges are used to transport components from the port. The strategy represented by the vessel affects both the size of the loading sets and the time it takes to install a turbine. Amorosi et al. (2024) consider the installation of foundation, cables, transition piece and turbine as separate installation activities, that need to be done by different vessels. The only paper focusing on vessel scheduling that allows each vessel to perform multiple installation activities is Irawan et al. (2017a).

Several of these papers also consider the question of what vessels should be chartered to perform the installation, and/or when, and for how long, to charter them. This leads to variants of fleet size and mix (FSM) problems for installing an OWF. Backe and Haugland (2017) present a strategic model that chooses both the number of vessels, the installation strategy, and the staging port(s) to use. Amorosi et al. (2024) consider the start- and end-week for the chartering of each vessel as a decision, and the objective is to minimize the total charting costs, and the lost electricity production. They consider that the OWF may start producing electricity as soon as all turbines connected to a given intra-array cable are installed, and take this into account when penalizing production losses.

When modeling the problem of scheduling multiple vessels, it is not sufficient to expand the model presented in Section 4.1 by simply adding a vessel index to the variables, since there is no way to ensure that the components are installed in the correct order. Amorosi et al. (2024) consider a special case where each component can only be installed by one vessel, and present a model where they keep track of the time when each component is installed on each turbine. This can easily be extended to the case where each vessel can install multiple components by creating a graph where each node represents a combination of a turbine and a component (i, k) for all $i \in \{1, \dots, C\}$ and $k \in \mathcal{K}$. The problem may then be modeled using a variable $x_{vikjk'}$ which is equal to one if vessel v installs component k' on turbine j directly after installing component k on turbine i . The times when components k and k' are installed on turbines i and j , respectively, may then be represented by variables t_{ik} and $t_{jk'}$, and the variables may be connected as follows:

$$t_{ik} + B_k \leq t_{i(k+1)}, \quad k \in \mathcal{K} \setminus \{|\mathcal{K}|\}, i \in \{1, \dots, C\}, \quad (43)$$

$$(t_{ki} + T_{vij} - t_{k'j})x_{vikjk'} \leq 0, \quad k, k' \in \mathcal{K}, i, j \in \{1, \dots, C\}. \quad (44)$$

An additional advantage of this model is that it is able to capture the time it takes to move the vessel between turbines through the parameter T_{vij} . If $i = j$ the vessel is installing a new component on the same turbine, and thus there is no need to re-position, jack-up/down etc., and T_{vij} can be given values accordingly. This is not possible to capture with the model given in Section 4.1, where all turbines are assumed to be at the same location.

An alternative approach, taken by Ait-Alla et al. (2013), Irawan et al. (2017a), and Rippel et al. (2019), is to partition the planning horizon into a set of discrete time steps, $\mathcal{T}$, which makes it possible to introduce a variable that counts the number of each component that has been installed in each time step. Using a time discrete model can also be used to handle weather limits implicitly by fixing variables stating that a vessel is installing a component at a given time step to zero for time steps when the weather conditions are above the limit for that component/vessel.

Another advantage of the discrete time model is that it is possible to use the Dantzig-Wolfe decomposition to create an alternative model where each variable represents a round trip r starting in time period t for vessel v . Formally, we define a bounded set of feasible solutions P_{vt} described by the subset of constraints (26)–(41), given that $\mathcal{I} = \{1\}$, $t_1^S = t$, and where the parameters B_k , W_p^S and W_p^E are set according to the specifications of vessel v , which we denote B_{kv} , W_{vp}^S and W_{vp}^E . Further, let Ω_{vt} be the set of extreme points of the convex hull of P_{vt} , i.e. the set of round trips. Note that as long as all time parameters have integral values, all t -variables will also have integral values in an extreme point. Let $L_{kv\omega}$ be the loading set used on round trip ω and let $T_{v\omega\tau}$ be equal to one if round trip ω is ongoing in time period $\tau \in \mathcal{T}$. Furthermore, let $A_{kv\omega\tau}$, and $B_{kv\omega\tau}$ be one if the installation of component k is started, or finished, in time period τ , and zero otherwise. To reduce the number of indices, we let $\Omega_v = \bigcup_{t \in \mathcal{T}} \Omega_{vt}$.

In addition, we let C_v^{TC} denote the day rate of chartering vessel v , and C^P be a penalty cost per day the wind farm is not finished. This penalty cost is usually based on the lost revenue from not producing and selling electricity. Using this notation, we may formulate a mathematical model in which the variable $\lambda_{v\omega}$ is equal to 1 if vessel v uses round trip $\omega \in \Omega_v$, and 0 otherwise. Furthermore, let the variable e denote the time the OWF installation is completed, and let variables s_v and e_v denote the start and end time of the charter period for vessel v .

$$\min \sum_{v \in \mathcal{V}} C_v^{TC} (e_v - s_v) + C^P e, \quad (45)$$

$$\sum_{v \in \mathcal{V}} \sum_{t \in \mathcal{T}} \sum_{\omega \in \Omega_v} L_{kv\omega} \lambda_{v\omega} = C, \quad k \in \mathcal{K}, \quad (46)$$

$$s_v \leq |\mathcal{T}| - (|\mathcal{T}| - t) \sum_{\omega \in \Omega_v} T_{v\omega t} \lambda_{v\omega}, \quad v \in \mathcal{V}, t \in \mathcal{T}, \quad (47)$$

$$t \sum_{\omega \in \Omega_v} T_{v\omega t} \lambda_{v\omega} \leq e_v, \quad v \in \mathcal{V}, t \in \mathcal{T}, \quad (48)$$

$$e_v \leq e, \quad v \in \mathcal{V}, \quad (49)$$

$$\sum_{\omega \in \Omega_v} T_{v\omega t} \lambda_{v\omega} \leq 1, \quad v \in \mathcal{V}, t \in \mathcal{T}, \quad (50)$$

$$\sum_{v \in \mathcal{V}} \sum_{\omega \in \Omega_v} \sum_{t=1}^{\tau} A_{(k+1)v\omega t} \lambda_{v\omega} \leq \sum_{v \in \mathcal{V}} \sum_{\omega \in \Omega_v} \sum_{t=1}^{\tau} B_{kv\omega t} \lambda_{v\omega}, \quad k \in \mathcal{K} \setminus \{|\mathcal{K}|\}, \tau \in \mathcal{T}, \quad (51)$$

$$\lambda_{v\omega} \in \{0, 1\}, \quad v \in \mathcal{V}, \omega \in \Omega_v. \quad (52)$$

The objective function (45) consists of two conflicting terms. The first is to minimize the total chartering costs of all vessels, while the second term minimizes the makespan of the installation process. Constraints (46) state that the number of times each component is installed must be equal to the number of turbines at the OWF, while constraints (47) and (48) set the start- and end-time of the charter of each vessel, and constraints (49) set the end time of the project. Furthermore, constraints (50) ensure that only one round trip is undertaken in each time period, while constraints (51) ensure that the number of components $k+1$ that have begun to be installed is smaller than the number of components k that have completed installation in each time period, thus ensuring that components are installed in the correct order. Finally, constraints (52) impose binary restrictions on the λ -variables.

The main advantages of this model is that most of the constraints, such as the weather conditions and the sequence in which components are installed on a single round trip, are implicitly handled when creating the variables of the problem. Consequently, the model itself contains relatively few constraints, and has a stronger LP-relaxation. The main disadvantage of the model is that the size of the set Ω_{vt} grows exponentially with the sizes of $\mathcal{K}$ and $\mathcal{J}$, making it impossible to generate all extreme points for large problem instances.

4.3. Project scheduling

Looking at the installation of an OWF from a strategic perspective, it is natural to consider it as a project, where a set of installation activities must be performed by a set of resources. Barlow et al. (2017) and Kerkhove and Vanhoucke (2017) present optimization problems related to the installation of OWFs, inspired by project scheduling, rather than job shop scheduling, models. In these models, certain details regarding the non-installation activities of the vessels, e.g. return trips to port and loading, are omitted, and the vessels are considered resources that are to be assigned to activities. Barlow et al. (2017) heuristically pre-assign a set of installation activities to a set of vessels, and solves a model based on *program evaluation review technique* (PERT) to handle precedence both between activities at each turbine and activities performed by each vessel. The model further operates with earliest/latest start/end-times for some of the tasks, e.g. earliest time a vessel is available, or latest time a certain turbine should be finished. The model finds the critical path through this precedence network. The model is used as part of a simulation framework that is used to establish the duration and cost of an installation project.

Kerkhove and Vanhoucke (2017) present a more high-level model to plan the entire installation project. The problem differs in that the main decision variable is when each installation activity should be ready to begin, which they refer to as *gate times*. They present a time discrete model, where vessels are considered resources, and the decisions are how many vessels of each type to charter in each time period. Unlike the other models presented, they allow for the possibility to temporarily decommission a vessel. The time is discretized into months, rather than days or hours, and they use a *net present value* approach in the objective function, due to the long planning horizon considered.

4.4. Handling uncertainty in the installation phase

The main source of uncertainty when planning installation activities at an offshore wind farm is the weather conditions, where the wind and waves put restrictions on what installation activities can be done at any given time. When doing short term planning for the next couple of days it might be reasonable to assume that the weather conditions are known, however, for longer planning horizons this uncertainty needs to be accounted for, to avoid unrealistically short installation times and charter periods.

Among those papers that handle uncertainty, we distinguish between those that handle it within the mathematical model, and those who handle it outside of it. Among the former we find Ursavas (2017) who present a two-stage stochastic programming model where the weather conditions are subject to uncertainty, while Barlow et al. (2017) and Amorosi et al. (2024) use a robustness budget (Bertsimas and Sim, 2004) and a *light robust* approach (Fischetti and Monaci, 2009), respectively, to handle uncertainty.

In the latter category, we find papers that integrate mathematical models with simulation models to account for the weather uncertainty. This is the case for Kerkhove and Vanhoucke (2017), Bae and Ko (2023), Barlow et al. (2017), and Rippel et al. (2019). Rippel et al. (2019) and Barlow et al. (2017) simulate weather conditions and use these simulations to determine the values of the duration parameter for different activities in their optimization model. On the other hand, Kerkhove and Vanhoucke (2017) use a simulation model to evaluate the solutions from their heuristic optimization procedures. Every time a heuristic generates a candidate installation schedule, it is evaluated by a simulation model to identify the expected time and cost of the schedule.

Finally, Bae and Ko (2023) develop a simulation model for the installation of an offshore wind farm, where the installation process is simulated multiple times using randomly generated weather data, to capture the uncertainty of the weather. An optimization model is used as a sub-routine to decide what components a given vessel should install on its next round trip, given the simulated weather for the coming days.

4.5. Solution methods

Due to the relatively few papers on problems studied in the installation phase of the OWF life cycle, we discuss the solution approaches jointly in this section. A complete overview can be found in Table 4.

Both exact and heuristic solution methods have been applied to problems in the installation phase. Among exact methods, solving the MIP model using a commercial solver is the most common (Scholz-Reiter et al., 2010; Ait-Alla et al., 2013; Backe and Haugland, 2017; Rippel et al., 2019; Bae and Ko, 2023; Amorosi et al., 2024). Two papers use decomposition strategies, where Ursavas (2017) apply Benders decomposition to solve the LP-relaxation of her two-stage stochastic program, while Irawan et al. (2017a) propose a model based on Dantzig-Wolfe decomposition, where all extreme points are generated a-priori.

The metaheuristic solution methods that have been used are a variable neighborhood search heuristic (Irawan et al., 2017a) to solve the multi-vessel scheduling problem, and a genetic algorithm (Kerkhove and Vanhoucke, 2017) to solve the project scheduling problem. Matheuristic approaches are proposed by Amorosi et al. (2024), where the original model is simplified in several ways, both to obtain primal and dual bounds on the optimal solution.

The literature on the installation phase thus far has focused mainly on developing richer problems and models. As a result of this, no two paper solves exactly the same problem, making comparisons of solution methods difficult. However, the fact that the problem shares many similarities with the job shop- and project scheduling problems indicate that methods suitable for solving these problems, are likely to work well also for problems in the installation phase of an OWF.

Paper	Planning level	Objective	#vessels	act. per v	Time	Model type	Uncertainty	solution method	# Turbines	components
Scholz-Reiter et al. (2010)	T	T	1	M	C	JS&PP	No	MIP-model	12	foundation, turbine
Ait-Alla et al. (2013)	T	C	M	M	D		No	MIP-model	30	foundation, pile, cable, turbine
Ursavas (2017)	T	T	1	M	C	JS&PP,2-SP,BD	weather - SP	Benders	16-80	foundation, turbine
Kerkhove and Vanhoucke (2017)	S	C	M	M	C	PS	weather	GA	54	foundation, scour protection, turbine (different configs.)
Backe and Haugland (2017)	S	C	M	1	C	JS&PP	No	MIP-model	20-100	turbine
Irawan et al. (2017a)	T	T & C	M	M	D	JS&PP,DWD	No	MIP-model, VNS	12-50	foundation, cables, turbine
Barlow et al. (2017)	S	T	M	M	C	PERT	weather/duration - RO	MIP-model	120	turbine
Rippel et al. (2019)	O	C	M	1	D	JS	No	MIP-model		turbine
Bae and Ko (2023)	O	T	1	M	C	JS	No	MIP-model	90-250	foundation, tower, turbine
Amorosi et al. (2024)	S	C	M	1	C	NF	weather - light RO	MIP-model, Matheuristic	49-72	foundation, cables, transition piece, turbine

Table 4: Overview of selected papers for the installation phase, ordered by publication year. **Planning level** Strategic (S), Tactical (T) or Operational (O); **Objective** quantity to be minimized, either time (T) or cost (C); **# vessels** single vessel (1) or multiple vessels (M); **act. per v** 1 if all vessels can only install one component, M if at least one vessel can install multiple components; **Model type** Job shop scheduling (JS), production planning (PP), project scheduling (PS), PERT, or network-flow (NF). DWD/BD added if decomposed using Dantzig-Wolfe or Benders decomposition is used; **Uncertainty** The sources of uncertainty in the model, and how it is captured SP (stochastic programming), RO (Robust optimization); **Solution method** (MIP, matheuristic, Benders Decomposition, VNS (variable neighborhood search), GS (Genetic algorithm); **# Turbines** The number of turbines included in the test instances; **Components** The components considered in the solution method/instances.

4.6. Data instances

Most of the data instances used in the surveyed papers use synthetic data instances, however Ursavas (2017) (*Bard 1* and *Borkum West*), Kerkhove and Vanhoucke (2017) (*Thorntornbank*), Bae and Ko (2023) (*Southwest*) and Amorosi et al. (2024) (*Horns Rev 3* and *Sandbank*) base their instances on real wind farms. With the exception of the first two, all articles reviewed in Table 4 consider a reasonable number of turbines to be installed. In the last decade, the main focus in the OWF industry has been to increase the size of the turbines, thus the number of OWTs per OWF has remained largely unchanged.

When it comes to the components considered in the data instances, all papers consider the turbine itself, while most also consider the foundation, and three papers consider the cables. The cables are installed using a special type of cable-laying vessel, and it could therefore be argued that if the goal is to minimize cost, cable-laying may be treated separately from the other two components. Several papers, also divide the installation of foundations into several sub-components, either including the *piles* that secure the foundation to the seabed, the *scour protection* added to the foundations after they are installed, or the *transition piece* that connects the foundation to the turbine itself.

As mentioned above, the turbine also consists of four sub-components, however, with the exception of Bae and Ko (2023) who consider the *tower* as a separate component, all papers treat the turbine as a single component, implying that the sub-components are installed together. Installing an OWT take anywhere from two to six lifting operations, depending on the installation strategy. Requiring a single weather window to span the duration of all those lifts likely leads to overly conservative models.

Another interesting observation from studying the input data used in different papers is that although several papers present models that allow multiple vessels to perform multiple tasks, only Ait-Alla et al. (2013) have several vessel types capable of installing the same component. The *mix* part of the FSM problems is thus not really captured in most of the data instances presented in the literature, as all vessels types are required to install the OWF.

5. Operations and maintenance

With an estimated operational OWF lifetime of 25–30 years, the operations and maintenance (O&M) phase lasts the longest. There are two branches of optimization problems in this phase: One is centered around OWF control with the objective of maximizing the overall electricity production by configuring the individual OWTs in the OWF. Formulating these problems requires detailed technical knowledge at a component level, and we therefore omit literature connected to this branch in this survey and refer to Andersson et al. (2021) for an overview.

The other branch has maintenance planning at its core. Even with continuous maintenance, it is approximated that each OWT fails on average 8.3 times per year (Carroll et al., 2016). This leads to *downtime*, where the OWTs do not produce electricity. Therefore, maintenance planning is a cornerstone in reducing the LCOE. This section is dedicated to a variety of interesting problems that occur in this context. Although they cover different aspects of O&M, they all share the goal of minimizing the overall cost by balancing *direct O&M costs* with *downtime costs* as illustrated in Figure 5.

Availability is a quantification of an OWT's (or an OWF's) readiness to produce electricity and can be given either as a percentage of time, of the potential electricity production, or of the electricity value. Direct O&M cost include all costs related to vessel chartering, new components for replacements, equipment, technician salary, etc. Downtime cost is the value of the electricity production that is lost when an OWT is stopped due to failure and/or as maintenance is carried out (Conroy et al., 2011). Hence, the magnitude of the downtime cost is highly dependent on whether maintenance is carried out before or after a failure has occurred. Using this as a discriminator, maintenance can be divided into two main categories: *Preventive maintenance* (PM) means that a maintenance task (MT) is executed *before* a failure occurs – to prevent it. Conversely, *corrective maintenance* (CM) means that an MT is executed *after* a failure has occurred – to correct it. For PM tasks, the downtime cost is calculated from the maintenance starts until it ends, while for CM tasks, the downtime cost is calculated from the failure occurs until the maintenance ends.

Vessels are crucial resources for the transportation of technicians, spare parts, and equipment to the OWF from the *O&M base(s)* and between the individual OWTs within the OWF. An O&M base is usually a port designated to service the OWF(s). The distance between an OWF and the shore is

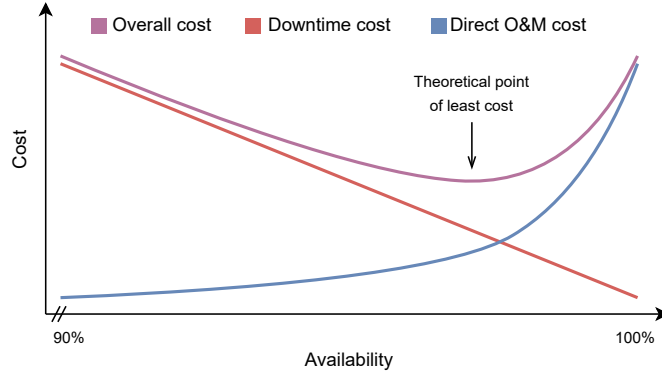


Figure 5: Balancing direct cost of O&M with downtime cost, based upon Philips et al. (2018)

typically 10-100 km (IRENA, 2025), while distances between the individual OWTs within an OWF are typically 650-1200 m (for currently commissioned OWFs) corresponding to eight times the OWT rotor diameter (Hernando et al., 2023). To reduce the distance from the O&M base to the OWF, an offshore platform or a mother vessel could alternatively serve as O&M base.

Various vessel types are used depending on the MT. The weather conditions govern the operability of the different vessel types. *Crew transfer vessels* (CTVs) are relatively small vessels that are used to transport a small group of technicians to the OWTs for minor repairs. *Service operation vessels* (SOVs) (also sometimes called *accommodation vessels*) are larger vessels that can host a larger group of technicians for longer time periods – typically 2-4 weeks – as they perform repairs, and can operate in tougher weather conditions compared to the smaller CTVs. Performing *heavy maintenance*, where larger components such as blades, generator, etc. are replaced, requires a *heavy lift vessel* (HLV) with a crane on board that is sensitive to high wind speeds. Typically, these operations are performed by a *jack-up vessel* (see Figure 2a), although there are other alternatives. Recently, *mother vessel* concepts have also been investigated in connection with O&M at OWFs. These are larger vessels that provide an offshore O&M base that can accommodate technicians and from which they can operate smaller CTVs, referred to as *daughter vessels*. In some cases, the mother vessels themselves can also be used to transport the technicians for minor repairs at the OWTs.

We adopt a classification scheme according to the length of the planning horizon as proposed by Shafiee (2015), beginning with the *operational level* in Section 5.1 treating *MT scheduling* problems. These problems have a short-term horizon, which can imply week-to-week or day-to-day decisions. It is worth noting that scheduling problems often occur as subproblems in *maintenance fleet size and mix* problems presented in Section 5.2. These problems can be considered on both a *tactical* and a *strategic level*. Tactical problems entail medium-term decisions, whose time horizon typically spans between half a year and five years, while strategic problems involve long-term decisions affecting the entire operational lifetime of an OWF. In Section 5.2 we discuss how uncertainty can be handled in the different models at operational, tactical, and strategic levels. At last we give an overview of the data instances used in the surveyed literature in Section 5.4

5.1. Operational problems

The OWF maintenance scheduling problems can be divided into two: the pure scheduling problem, which is considered in Section 5.1.1, and the combined routing and scheduling problem, which is considered in Section 5.1.2. The latter can be seen as a finer-granulated version of the former, as the routing of the vessels, and subsequently the traveled distances, are taken into account.

5.1.1. Scheduling

In the WF maintenance scheduling problem (WFMSPP), the goal is to decide which MTs to perform when and to allocate *resources*. In this context, resources can be different types of vessels and/or technicians, as well as necessary spare parts. The time horizon typically ranging from a few days up to one week.

When considering such a short time horizon, both weather conditions and component degradation can be assumed known. This means that if we consider a pure version of the problem, stripped from details regarding vessel/vehicle capabilities, technicians' shift regulations, and other fine-grained specifications, the offshore and onshore WFMSPs are essentially the same. Although the literature treating maintenance scheduling problems for electricity systems is, in general, rich (Froger et al., 2016), the attention that the WFMSP has received is limited - both when it comes to offshore and onshore WFs.

Kovács et al. (2011) present a detailed onshore WFMSP model considering multiple maintenance teams servicing multiple WFs. They apply their model in a rolling horizon framework spanning 3–7 days, where the plan is updated each day, and only the first day in the schedule is actually executed. Below, we present a simplified version of their model adapted for the service of one OWF. Even though the model itself is quite simple – and possible to solve for real-world instances – we include this formulation of the problem, as it plays an important role as a subproblem in many of the problems discussed in the following sections.

Starting with the notation, let $\mathcal{N}$ be a set of MTs in the OWFs that are to be assigned to a set of vessels $\mathcal{V}$ in a *scheduling period*, where the time horizon is discretized into a set of time periods $\mathcal{T}$. We define a parameter D_i as the duration of MT $i \in \mathcal{N}$ and P_v^T as the number of technician teams on board vessel $v \in \mathcal{V}$. Another parameter A_{vt} is defined to be 1 if vessel $v \in \mathcal{V}$ is available and able to perform maintenance in time period $t \in \mathcal{T}$, and 0 otherwise. This can, e.g., be used to model weather conditions and their impact on the different vessels.

Each MT $i \in \mathcal{N}$ is associated with a downtime cost C_{it}^D depending on which time period $t \in \mathcal{T}$ it is started. If the MT is started before the failure occurs (requiring the OWT to shut down), C_{it}^D entails only the lost production cost due to the turbine being shut down in the maintenance period. If the opposite is the case, C_{it}^D in addition entails the lost production cost in the time period from the failure occurs and until the MT is started. If an MT is not assigned to a vessel within the scheduling horizon, it is postponed (to a later scheduling period outside the model). This is associated with a penalty cost C_i^P . We define the decision variable x_{ivt} to be 1 if MT $i \in \mathcal{N}$ is started by a technician team on board vessel $v \in \mathcal{V}$ in time period $t \in \mathcal{T}$, and 0 otherwise. In addition, we define an auxiliary variable y_i to be 1 if an MT is postponed, and 0 otherwise.

$$\min \sum_{i \in \mathcal{N}} \sum_{v \in \mathcal{V}} \sum_{t \in \mathcal{T}} C_{it}^D x_{ivt} + \sum_{i \in \mathcal{N}} C_i^P y_i, \quad (53)$$

$$\text{st. } \sum_{v \in \mathcal{V}} \sum_{t \in \mathcal{T}} x_{ivt} + y_i = 1, \quad i \in \mathcal{N}, \quad (54)$$

$$D_i x_{ivt} \leq \sum_{t'=t}^{t+D_i-1} A_{vt'}, \quad i \in \mathcal{N}, v \in \mathcal{V}, t \in \mathcal{T}, \quad (55)$$

$$\sum_{i \in \mathcal{N}} \sum_{t'=t-D_i+1}^t x_{ivt'} \leq P_v^T, \quad v \in \mathcal{V}, t \in \mathcal{T}, \quad (56)$$

$$x_{ivt} \in \{0, 1\}, \quad i \in \mathcal{N}, v \in \mathcal{V}, t \in \mathcal{T}, \quad (57)$$

$$y_i \in \{0, 1\}, \quad i \in \mathcal{N}. \quad (58)$$

The objective function (53) minimizes the downtime cost. Constraints (54) ensure that each MT is either performed or penalized in the objective function. The next constraints (55) state that an MT can only be assigned to a vessel if the vessel is available for the full duration of the task, while constraints (56) ensure that each vessel is not assigned more MTs in parallel than the number of technician teams on board. Lastly, constraints (57) and (58) establish the domain of the variables.

The main advantage of this model is that, although the number of constraints and variables grows polynomially with the input size of $\mathcal{N}$, $\mathcal{V}$, and $\mathcal{T}$, it remains computationally tractable for realistic instances. In practice, this is due to both $\mathcal{N}$ and $\mathcal{V}$ being relatively small, and given an operational time horizon, with an appropriate granulation of time, the same applies to $\mathcal{T}$. Further, the constraints consist of one set of set-partitioning constraints (54) and two sets of knapsack constraints (55)-(56), both of which

are well researched (see, e.g., Hoffman and Padberg (2024) and Cacchiani et al. (2022), respectively). A major drawback of this model is its inability to capture routing between the O&M base(s) and OWF(s), as well as between OWTs within the OWF(s). Considering the short distances between OWTs, this may be a reasonable approximation when the WFMSp acts as a subproblem in a tactical or strategic setting and when only one OWF is considered, but it will otherwise require some alterations.

Besnard et al. (2009) was the first to formulate an offshore WFMSp model. They investigated an opportunistic maintenance strategy by including optional preventive MTs, which can be performed when a vessel with technicians is visiting the OWF in connection with corrective MTs. The objective function accounts for transportation costs, downtime costs, and overtime pay for the technicians.

Despite their focus on onshore WFs, it is worth mentioning a few recent articles: Froger et al. (2018) present a WFMSp model capturing fine-grained details including task assignment to a multi-skilled workforce and propose a constraint programming-based *large neighborhood search* (LNS) to solve it. Froger et al. (2017) solved the same problem with an exact method using a branch-and-check approach that utilized both Benders cuts and problem-specific cuts. Finally, Pérez (2022) presents an online scheduling framework in which the WFMSp is repeatedly solved by a GA as part of a simulation model with a strategic time horizon.

5.1.2. Routing and scheduling

The OWF maintenance routing and scheduling problem (WFMRSp) consists of assigning a heterogeneous fleet of vessels to support MTs at an OWF, while considering the routing of the vessels. Each vessel starts its route at (one of) the O&M base(s), travels to the OWF(s), where the OWTs requiring maintenance are visited, before the vessel returns to (one of) the O&M base(s). Similarly to the WFMSp, the time horizon is typically between one day and up to a week. The short time horizon makes it sensible to consider the routing despite the short distances between OWTs in the OWF(s).

Contrary to the WFMSp, the WFMRSp is more well visited in the literature. In Table 5 we present an overview of characteristics of selected literature presenting mathematical models for the offshore WFMRSp. Dai et al. (2015) were the first to formulate an optimization model for the offshore WFMRSp. Like the formulations in the majority of the reviewed papers published later, their formulation can be classified as a *vehicle routing problem with pickup and delivery* (VRPPD) in accordance with Berbeglia et al. (2007). However, in the WFMRSp technicians are delivered at the OWTs along with the necessary equipment to perform an MT(s), and are picked up again *after* the MT(s) are completed. This reverse order of deliveries and pickups differentiates it from traditional VRPPD together with unique features where: i) the fleet consists of heterogeneous vessels with capacities on technicians and equipment, ii) the vessel operability is governed by the weather conditions, and iii) there are individual penalties associated with delaying the visit to each node. Furthermore, it is often multi-period to capture time horizons exceeding one day.

Here, we use a simplified version of the formulation in Dai et al. (2015) that has been restricted to a time horizon of one day to demonstrate the model's features. Equivalent to the notation for the WFMSp, we have a set of vessels $\mathcal{V}$ that are assigned to n MTs. The routing in the WFMRSp can be captured by formulating the problem on a graph $G = (\mathcal{N}, \mathcal{A})$. Let each of the n MTs be associated with two nodes: a delivery node represents the delivery of technicians along with the necessary equipment and spare parts to perform the MT on the OWT, while a pickup node represents picking up technicians after the MT is performed. Define $\mathcal{D} = \{1, \dots, n\}$ to be the set of delivery nodes, and $\mathcal{P} = \{n+1, \dots, 2n\}$ to be the set of pickup nodes. Each MT i then consists of a delivery node $i \in \mathcal{D}$ and a pickup node $n+i \in \mathcal{P}$. Note that when convenient, we can let the set of delivery nodes $\mathcal{D}$ represent the set of MTs. Furthermore, let $\{0, 2n+1\}$ be the set of nodes representing the O&M base where each vessel begins and ends its route. Hence, in the graph G , the set of nodes $\mathcal{N}$ consists of the union set $\mathcal{D} \cup \mathcal{P} \cup \{0, 2n+1\}$, while the set of arcs $\mathcal{A}$ consists of pairs (i, j) such that $\mathcal{A} = \{(i, j) | i, j \in \mathcal{N}, i \neq j, j \neq i-n, j \neq 0, i \neq 2n+1\}$.

Paper	Objective	Extensions				MT types		Uncertainty	Model type	Solution Method
		#periods	#WFs	#bases	#tech types	PM	CM			
Dai et al. (2015)	Tr + P	M	1	1	1	×		-	VRPPD	MIP
Zhang (2014)	Tr + P	M	1	1	1	×		-	VRPPD	Duo-ACO
Stålhane et al. (2015)	Tr + P + D	1	1	1	1	×	×	-	SPP + SPPRC	DW decomp., MIP + HLA
Irawan et al. (2017b)	Tr + P + L	M	M	M	M	×		-	SPP + SPPRC	DW decomp., MIP + A priori
Raknes et al. (2017)	Tr + P + D	M	M	1	1	×	×	-	VRPPD	Rolling horizon heuristic
Schrotenboer et al. (2018)	Tr + P + L	M	M	M	M	×		-	SPP	Two-stage ALNS
Schrotenboer et al. (2019)	Agg + L	M	M	M	M	×		-	SCP + SPPRC	DW decomp., Branch&Price&Cut
Irawan et al. (2021)	Tr + P + D	1	1	1	1	×	×	weather - SP	VRPPD	LNS + MCS
Irawan et al. (2022a)	Tr + P + D	1/M	1	1	M	×	×	(weather)	SP + VRPPD	MIP + Matheuristic incorp. VNS
O’Neil et al. (2023)	Tr + P + D + L	1/M	1	1	M	×	×	-	SPP + SMP&OP	DW decomp., CG with HLA
Ma et al. (2024)	Ti + P	1	1	1	M	×	×	-	VRPPD	Matheuristic
Feng et al. (2024)	Tr + P + D	M	M	1	1	×		weather - DRO	VRPPD	Column-and-constraint algorithm
Li et al. (2024)	Ti + P + D + L + S + E	M	1	1	1	×			VRP	CQHHH
Si et al. (2025)	Tr + Ti + P + D	1/M	1	1	1	×	×	(weather)	SP + TSPPD	Enumeration + GA

Table 5: Overview of selected papers for the WFMRSPP, ordered by publication year. **Objective** is the quantity to be minimized, can be travel cost (Tr), completion time (Ti), penalty (P), downtime cost (D), labor cost (L), aggregated cost (Agg), safety (S) and emissions (E). **Extensions** common extensions where **#periods** is either single period (1) or multiple periods (M), **#OWFs** is either single OWF (1) or multiple OWFs (M), **#bases** is either single O&M base (1) or multiple O&M bases (M), and **#tech. types** is either single technician type (1) or multiple technician types (M). **MT types** is the maintenance task (MT) type according to our definitions of PM and CM in Section 5 which is **PM** preventive maintenance and **CM** corrective maintenance. **Uncertainty** is the source(s) of uncertainty in the model and how this is captured by either stochastic programming (SP) or distributionally robust optimization (DRO). **Model type** is either vehicle routing problem with pick up and delivery (VRPPD), set partitioning problem (SPP), shortest path problem with resource constraints (SPPRC), set covering problem (SCP), selective maintenance and orienteering problem (SM&OP), vehicle routing problem (VRP), scheduling problem (SP), or traveling salesperson with pick up and delivery (TSPPD). **Solution method** is large neighborhood search (LNS), duo ant colony optimization (Duo-ACO), chaotic quantum harris hawks heuristic (CQHHH), or monte carlo simulation (MCS).

Each MT $i \in \mathcal{D}$ has a duration of D_i and must be performed to avoid a penalty cost C_i^P . Let W_i be the weight of the equipment and spare parts needed for the MT, and let P_i be the number of technicians it requires to perform it. For convenience, we assume $P_{n+i} = -P_i$ for each pickup node $n+i \in \mathcal{P}$ and $P_0 = P_{2n+1} = 0$ for the ports. Each vessel $v \in \mathcal{V}$ has a weight capacity W_v^{MAX} on equipment and spare parts, and a personnel capacity of P_v^{MAX} technicians. As the weather conditions govern when each vessel can operate, a parameter T_v^{MAX} sets the maximum number of working hours at the OWF for each vessel $v \in \mathcal{V}$. Letting vessel $v \in \mathcal{V}$ traverse arc $(i, j) \in \mathcal{A}$ is associated with a travel cost C_{ijv} where $C_{i(n+i)v} = 0$, i.e., a MT's delivery node i and pickup node $n+i$ are visited consecutively by vessel v . We define four decision variables: Let the binary variable x_{ijv} be 1 if arc $(i, j) \in \mathcal{A}$ is traversed by vessel $v \in \mathcal{V}$, and 0 otherwise. Define the variable t_{iv} to be the time when node $i \in \mathcal{N}$ is visited by vessel $v \in \mathcal{V}$, while q_{iv} is the number of technicians on board vessel $v \in \mathcal{V}$ right after it leaves node $i \in \mathcal{N}$. With this notation, we formulate the WFMRS as follows:

$$\min \sum_{(i,j) \in \mathcal{A}} \sum_{v \in \mathcal{V}} C_{ijv} x_{ijv} + \sum_{i \in \mathcal{D}} C_i^P (1 - \sum_{j \in \mathcal{N}} \sum_{v \in \mathcal{V}} x_{ijv}), \quad (59)$$

$$\text{s.t. } \sum_{j \in \mathcal{N}} \sum_{v \in \mathcal{V}} x_{ijv} \leq 1, \quad i \in \mathcal{D} \cup \mathcal{P}, \quad (60)$$

$$\sum_{j \in \mathcal{N}} x_{0jv} = 1, \quad v \in \mathcal{V}, \quad (61)$$

$$\sum_{i \in \mathcal{N}} x_{i(2n+1)v} = 1, \quad v \in \mathcal{V}, \quad (62)$$

$$\sum_{j \in \mathcal{N}} x_{jiv} = \sum_{j \in \mathcal{N}} x_{ijv}, \quad i \in \mathcal{D} \cup \mathcal{P}, v \in \mathcal{V}, \quad (63)$$

$$\sum_{j \in \mathcal{N}} x_{ijv} = \sum_{j \in \mathcal{N}} x_{(n+i)jv}, \quad i \in \mathcal{D}, v \in \mathcal{V}, \quad (64)$$

$$t_{0v} = 0, \quad v \in \mathcal{V}, \quad (65)$$

$$t_{(2n+1)v} \leq T_v^{MAX}, \quad v \in \mathcal{V}, \quad (66)$$

$$t_{(n+i)v} - t_{iv} \geq D_i, \quad i \in \mathcal{D}, v \in \mathcal{V}, \quad (67)$$

$$(t_{iv} + T_{ijv} - t_{jv}) x_{ijv} \leq 0, \quad (i, j) \in \mathcal{A}, v \in \mathcal{V}, \quad (68)$$

$$\sum_{i \in \mathcal{D}} \sum_{j \in \mathcal{N}} W_i x_{ijv} \leq W_v^{MAX}, \quad v \in \mathcal{V}, \quad (69)$$

$$(q_{iv} - P_j - q_{jv}) x_{ijv} = 0, \quad (i, j) \in \mathcal{A}, v \in \mathcal{V}, \quad (70)$$

$$q_{iv} \leq P_v^{MAX}, \quad i \in \mathcal{N}, v \in \mathcal{V}, \quad (71)$$

$$t_{iv} \geq 0, \quad i \in \mathcal{N}, v \in \mathcal{V}, \quad (72)$$

$$q_{iv} \in \mathbb{Z}_0^+, \quad i \in \mathcal{N}, v \in \mathcal{V}, \quad (73)$$

$$x_{ijv} \in \{0, 1\}, \quad (i, j) \in \mathcal{A}, v \in \mathcal{V}. \quad (74)$$

The objective function (59) minimizes the total travel costs of all the vessels and the penalty costs associated with not performing maintenance tasks. The first term minimizing the travel costs gives the model an incentive to schedule MTs at OWFs sited near each other consecutively, while the second term minimizing the penalty costs gives the model an incentive to schedule MTs as early as possible. Note that the travel costs in this formulation are based on distance, but could alternatively have been based on time. Constraints (60)-(64) are related to vessel routing. Constraints (60) ensure that each delivery and pickup node is visited at most once. Constraints (61) and (62) guarantee that each vessel begins and ends its route in the O&M base, respectively, while the flow between the other nodes is conserved by constraints (63).

Constraints (64) express that if the technicians are delivered to perform an MT at an OWT, then they must be picked up by the same vessel. Next, constraints (65)-(68) concern maintenance scheduling.

Constraints (65) initialize the schedule by setting the departure time to 0, while constraints (66) ensure that the vessel returns to the O&M base within the maximum number of working hours. Constraints (67) guarantee that the technicians are picked up after the MT is performed. Constraints (68) account for the travel time of each vessel in the schedule. Constraints (69) ensure that the weight of the loaded spare parts and equipment respects the load capacity of the vessel. Constraints (70) keep track of the number of technicians on board the vessel, while constraints (71) ensure that the personnel capacity of the vessel is respected. Lastly, constraints (72)-(74) establish the domain of the decision variables.

The main advantages of this model are that it incorporates routing between the OWF(s) and O&M base(s) as well as between OWTs within the OWF(s), and that it is based on the well-known VRPPD problem, for which a large body of research on solution methods exists. However, due to constraints (68) and (70), which require linearization using the big- M method, the LP relaxation of the VRPPD is known to be weak (Battarra et al., 2014), making it time-consuming to solve even for relatively small instances of the problem to optimality using commercial MIP solvers (see Dai et al. (2015)).

The objective function given by (59), where travel costs are balanced with penalty costs, is the most commonly used in the literature. However, other WFMRSR models include additional factors in their objective functions: safety and emissions (Purvis et al., 2019), labor costs (Schrotenboer et al., 2019; O’Neil et al., 2023), and completion time (Ma et al., 2024; Si et al., 2025).

The most common extension of the WFMRSR in the literature (see Table 5) is to consider multiple time periods. Considering a planning period that is discretized into a set of time periods, e.g., days, $\mathcal{T}$, we can add the index $t \in \mathcal{T}$ to all three variables (q_{ivt} , t_{ivt} , and x_{ijvt}). In addition, we add a sum over $t \in \mathcal{T}$ in the objective function (59) and on the left-hand side of constraints (60). Furthermore, each set of constraints (61)-(74) is expanded with a dimension for all $t \in \mathcal{T}$. In the objective function (59), we also introduce the parameter C_{it}^D from the WFMSR, which is multiplied by x_{ijvt} and summed over all indices. Although this formulation assumes that the vessels must return to the O&M base by the end of each period, there are vessel types, such as SOVs, that can stay offshore for several periods. Raknes et al. (2017) formulate a model capturing the use of SOVs.

Irawan et al. (2022a) and Si et al. (2025) explore a longer planning horizon by splitting the WFMSR into two parts: a maintenance scheduling problem (MSP) with a tactical horizon of one month, solved under the assumption that information on failures and weather forecast is known, and a maintenance routing problem (MRP) with an operational horizon, solved daily using updated information on failure and weather forecasts.

There are also several extensions of the WFMRSR connected to the considered MTs. About half of the articles consider both PM and CM as defined in the introduction of this chapter. When considering CM tasks, the parameter C_{it}^P in the objective function usually represents the accumulated downtime cost incurred until the end of time period t . Another common extension related to MTs (see e.g., Raknes et al. (2017); Irawan et al. (2017b); Schrotenboer et al. (2018)) is the inclusion of tasks where the vessel is required to stay at the OWT as the MT is performed. This is typically relevant if the MT involves a hazard, e.g., diving inspections. One paper (Raknes et al., 2017) considers the possibility of splitting MTs over multiple days.

Some papers also consider the skills of the technicians in their models. In O’Neil et al. (2023) the time it takes to perform an MT and the labor cost depend on which maintenance team is chosen to perform the MT. Irawan et al. (2017b), Schrotenboer et al. (2018), Irawan et al. (2022a) and Ma et al. (2024) on the other hand, view the MTs on a more detailed level where technicians with different skills are required depending on the type of task. The WFMRSR formulation given above can be extended to capture this by altering the notation to include an extra set, technician types $\mathcal{L}$. Its corresponding index l is added to the decision variable q_{ivl} and the parameter P_{il} . Then a sum over $l \in \mathcal{L}$ is added to the left-hand side of constraints (71). To ensure that constraints (70) and (73) are enforced for each technician type, we expand them with a dimension for all $l \in \mathcal{L}$.

The first four of the aforementioned papers account for the technician labor cost in the objective function. In our WFMRSR model this can be achieved by introducing a parameter C_l^L that is the labor cost of each technician type l , and a third term added to the objective function (59) is formulated as $\sum_{v \in \mathcal{V}} \sum_{l \in \mathcal{L}} C_l^L q_{ovl}$. Building upon this, the model can be extended to include other details such as constraining the number of technicians available at the O&M base(s) (Irawan et al., 2017b, 2022a). Schrotenboer et al. (2018) extend this further by allowing the allocation of technicians to a set of O&M

bases be one of the decisions in the problem, thus formulating the technician allocation and routing problem which is a generalization of the WFMRSRSP.

Another interesting generalization that is studied by O’Neil et al. (2023) is the selective maintenance and routing problem. In this problem, the subset of OWTs to maintain as well as the level of maintenance to perform on these is considered based upon the health of the OWTs, where the health is defined in terms of the reliability of a number of subsystems in each OWT. This gives the problem a structure similar to the *team orienteering problem* (TOP), which can be classified as a *VRP with profit* (Archetti et al., 2014), rather than the VRPPD structure adopted in most papers. Feng et al. (2024) consider another generalization, where the power generation schedule is taken into account. An electricity demand is given, and then maintenance and power production must be planned such that both the deviation from the demand, and the maintenance cost, are minimized.

Finally, it is worth mentioning a variant of the problem which occurs when mother vessels along with their daughter vessels are included in the fleet, requiring their routes to be synchronized. This variant shares similarities with mother vessel and drone routing problems (Poikonen and Golden, 2017; Amorosi et al., 2023). So far only Irawan et al. (2022b) have studied this in relation to OWF maintenance.

Furthermore, two papers integrate their optimization model into a framework to simulate the impact under different strategic decisions. Raknes et al. (2017) investigate the impact of different settings for the fleet size and mix under uncertainty in failures and weather conditions. Schrottenboer et al. (2018), on the other hand, investigate the impact on technician allocation and maintenance routing under different allocation policies under uncertainty in the OWF layout in terms of distances between OWFs, between O&M bases, and between OWFs and O&M bases.

In the literature, both exact and heuristic methods have been used to solve the WFMRSRSP. Dai et al. (2015) solve the multi-period version of (59)–(74) directly using a commercial solver, while (Stålthane et al., 2015; Irawan et al., 2017b; Schrottenboer et al., 2019; O’Neil et al., 2023) decompose the problem, using Dantzig-Wolfe decomposition, by recognizing that constraints (60) couple together a set of $|\mathcal{V}|$ independent subproblems. The subproblem for each $v \in \mathcal{V}$ is defined by a bounded set of feasible solutions, described by constraints (61)–(74) for that particular v , where each solution represents a feasible path through G from node 0 to $2n + 1$. The problem can then be formulated as a set-partitioning problem, where one path is selected for each vessel v so that all nodes are visited once, and the total cost of the paths selected, is minimized.

Irawan et al. (2017b) use an enumeration strategy to generate all paths a priori, while Schrottenboer et al. (2019) apply a branch-and-price-and-cut algorithm, where the paths are generated dynamically, through column generation. They further propose new resource-exceeding path inequalities to strengthen the formulation. Both Stålthane et al. (2015) and O’Neil et al. (2023) also rely on the decomposition approach described above, but generate paths heuristically. Stålthane et al. (2015) propose a heuristic dynamic programming algorithm to generate paths a priori, while O’Neil et al. (2023) employ a column generation approach in which paths are generated both by using commercial solvers and an adapted version of the dynamic programming algorithm proposed by Stålthane et al. (2015).

Several other heuristic methods have also been applied to solve the WFMSRP. Zhang (2014) propose duo ant colony optimization. Li et al. (2024) explore another bio-inspired method. Si et al. (2025) use an enumeration strategy using a Stirling diagram for their MSP, while employing a genetic algorithm to solve their MRP. Schrottenboer et al. (2018) propose a two-stage ALNS, Irawan et al. (2021) combine large neighborhood search with Monte Carlo simulation, while Irawan et al. (2022a) solve their MSP using exact methods as they develop a matheuristic integrating exact methods with variable neighborhood search to solve their MRP. Ma et al. (2024) also develop a matheuristic in which a heuristic assigns tasks to vessels and a commercial solver is used for the routing problem. Raknes et al. (2017) propose a rolling horizon heuristic, while Feng et al. (2024) pursue a column-and-constraint algorithm.

5.2. Tactical and strategic problems

The majority of the papers considering O&M for OWF at the tactical and strategic level focus on maritime fleet composition problems. Therefore, this section is dedicated to these. Before diving into them, it is worth highlighting another interesting problem occurring on the tactical level, which, to the best of our knowledge, has only been studied once for OWFs: Neves-Moreira et al. (2021) investigate optimal stock levels of components at SOVs – a variant of the repair kit problem (see e.g., Bijvank et al.

(2010)). To capture uncertainty in weather-dependent component degradation, they present a two-stage stochastic formulation. The first-stage model entails tactical decisions on stock levels. The second-stage model entails operational decisions on component deployment as OWTs are maintained in a formulation resembling a multi-period WFMRSPP without pickup nodes.

In a comprehensive overview, Pantuso et al. (2014) highlight that, in general, maritime fleet composition problems differ from their land-based counterparts by involving a higher level of uncertainty, a greater amount of capital, and the ships' value function which does not necessarily decrease over time. We believe it is reasonable to assume that this also applies to the fleet composition problems occurring within offshore wind. In the following, we survey the literature and present a model related to these.

In the offshore *WF maintenance fleet size and mix problem* (WFMFSMP), the goal is to select the composition of a heterogeneous fleet of vessels by choosing how many of each vessel type should be chartered (and/or bought) and when, so that the total costs, comprising the fleet investments and the operational costs of utilizing the fleet to perform MTs, are minimized. Furthermore, decisions regarding which O&M bases (onshore or offshore) to use, and the allocation of vessels to these, are also considered as a part of the problem. This characteristic distinguishes the WFMFSMP from other maritime fleet size and mix problems and is in common with facility location-routing problems, where strategic decisions on facility location and tactical decisions on vehicle routing are interdependent (Mara et al., 2021).

The WFMFSMP can be considered at either the strategic or tactical level. For strategic variants, the time horizon might span approximately 25 to 30 years, which is the design life of the OWT, while tactical variants typically span a few years. In both cases, operational decisions regarding fleet deployment are inherently part of the problem, that supports long-term decisions. The WFMFSMP has received some attention in the literature. One line of research focuses on simulation (e.g., van de Pieterman et al. (2011) and Dalgic et al. (2015a)). Another line of research focuses on optimization studies, and here, Halvorsen-Weare et al. (2013) was the first to introduce a mathematical formulation of the problem.

In Table 6, we present an overview of characteristics of selected literature utilizing optimization. Notice that all papers consider both CM and PM. This ensures that the long-term decisions on fleet size and mix, as well as use of O&M bases, are taken with full overview of the maintenance demand. Three of the papers consider the problem at a strategic level, and four at a tactical level. Most aspects of the modeling are similar for the two variants. Therefore, we present the commonalities in the following model inspired by Stålhane et al. (2019) and point out the differences where they occur.

Starting with the notation, let $\mathcal{K}$ be the set of onshore and/or offshore O&M bases, while $\mathcal{T}$ is the set of periods into which the time horizon is discretized. A typical time period is one month. Let $\mathcal{V}^T$ be the set of vessel types upon which we define the subset $\mathcal{V}_k^T \subseteq \mathcal{V}^T$ that is the set of vessel types that can operate from O&M base $k \in \mathcal{K}$. Operating from an O&M base $k \in \mathcal{K}$ is associated with an operating cost C_k^B . Each vessel $v \in \mathcal{V}$ is associated with a long-term charter cost C_v^{LT} , and a short-term charter cost C_{vm}^{ST} related to the charter period $m \in \mathcal{T}$. Define V_{vm}^{MAX} as the maximum number of vessels of type $v \in \mathcal{V}$ that can be chartered in time period $m \in \mathcal{T}$. Let B_{vk}^{MAX} be the maximum capacity at O&M base $k \in \mathcal{K}$ for vessel type $v \in \mathcal{V}_k^T$. Finally, we define three decision variables: Let β_k be one if O&M is used, and zero otherwise. γ_{kv}^{LT} is the number of long-term chartered vessels of type $v \in \mathcal{V}^T$ stationed at O&M base $k \in \mathcal{K}$, while γ_{kvm}^{ST} is the number of short-term chartered vessels of type $v \in \mathcal{V}^T$ stationed at O&M base $k \in \mathcal{K}$ in time period $m \in \mathcal{T}$. In addition, define γ^{LT} and γ^{ST} as a given set of long-term chartering decisions and short-term chartering decisions, respectively, and let $Q(\gamma^{LT}, \gamma^{ST})$ be a function estimating the expected cost of utilizing the fleet defined by γ^{LT} and γ^{ST} . The model is then formulated as follows:

Paper	Planning level	Objective	Features				MT types		Uncertainty	Model type	Solution Method
			MV	MPV	AB/MP	C out	PM	CM			
Halvorsen-Weare et al. (2013)	T	B + V + TR + P + D		×	×		×	×	-	det	MIP default solver
Gundegjerde et al. (2015)	T	1st: B + V, 2nd: V, 3rd: TR + P + D		×	×		×	×	W, F, EP, VSR	3-SP	MIP default solver
Stålthane et al. (2016)	S	1st: B + V + FV + R, 2nd: V + R + TR + D	×			×	×	×	W, F	2-SP	MIP default solver
Gutierrez-Alcoba et al. (2019)	T	1st: B + V, 2nd: TR + P + D		×	×		×	×	W, F	2-SP	MIP default solver + a priori
Stålthane et al. (2019)	T	1st: B + V, 2nd: TR + P + D	×	×	×		×	×	W, F	2-SP, DW	MIP solver + HLA
Schrotenboer et al. (2020)	T	1st: VA, 2nd: TR + PD + L			×		×	×	W, F	2-SP	SAA
Stålthane et al. (2021)	S	1st: B + V + R, 2nd: TR + P + D				×	×	×	W,F, EP, OWF dev	DL-SP	Integer L-shaped
Bolstad et al. (2022)	S	1st: B + V + R, 2nd: TR + P + D				×	×	×	W,F, EP, OWF dev	DL-SP	GRASP

Table 6: Overview of selected papers for the WFMFSMP, ordered by publication year. **Planning level** tactical (T) or strategic (S) **Objective** is the quantity to be minimized, can be base cost (B), cost of purchasing/chartering in vessels (V), fleet value (FV), revenue of chartering out/selling vessels (R), vessel assignment to base cost (VA), vessel travel cost (TR), penalty (P), penalty of vessel changing depot (PD) downtime cost (D), and labor cost (L). **Features** mother vessels (MV), vessels staying offshore for multiple periods (MPV), activity bundles/maintenance patterns (AB/MP), chartering out vessels (C out) **Uncertainty** weather (W), failures (F), electricity prices (EP), vessel spot rates (VSR) and/or OWF development (OWF dev) **Solution method** heuristic labeling algorithm (HLA), sample average approximation (SAA), or greedy randomized adaptive search procedure (GRASP).

$$\min \sum_{k \in \mathcal{K}} C_k^B \beta_k + \sum_{k \in \mathcal{K}} \sum_{v \in \mathcal{V}_k^T} C_v^{LT} \gamma_{kv}^{LT} + \sum_{k \in \mathcal{K}} \sum_{v \in \mathcal{V}_k^T} \sum_{m \in \mathcal{T}} C_{vm}^{ST} \gamma_{kvm}^{ST} + Q(\gamma_{kv}^{LT}, \gamma_{kvm}^{ST}), \quad (75)$$

$$s.t. \quad \sum_{k \in \mathcal{K}} \gamma_{kvm}^{ST} \leq V_{vm}^{MAX}, \quad v \in \mathcal{V}^T, m \in \mathcal{T}, \quad (76)$$

$$\gamma_{kv}^{LT} + \gamma_{kvm}^{ST} \leq B_{vk}^{MAX} \beta_k, \quad k \in \mathcal{K}, v \in \mathcal{V}_k^T, m \in \mathcal{T}, \quad (77)$$

$$\beta_k \in \{0, 1\}, \quad k \in \mathcal{K}, \quad (78)$$

$$\gamma_{kv} \in \mathbb{Z}_0^+, \quad k \in \mathcal{K}, v \in \mathcal{V}_k^T, \quad (79)$$

$$\gamma_{kvm} \in \mathbb{Z}_0^+, \quad k \in \mathcal{K}, v \in \mathcal{V}_k^T, m \in \mathcal{T}. \quad (80)$$

The objective function (75) minimizes the overall cost. The first term is the cost of using the O&M bases. The second and third terms are the long-term and short-term charter costs, respectively. The fourth term is the expected cost of utilizing the maintenance fleet. Constraints (76) set the capacity on short-term chartered vessels in each time period. Constraints (77) ensure that the vessel capacities of the O&M bases in use are respected. The last three constraints (78)-(80) establish the domain of the decision variables.

The main advantage of this model, similar to the WFMSP, is that despite the polynomial growth in the number of constraints and variables with respect to the input sets $\mathcal{K}$, $\mathcal{V}_k^T$, and $\mathcal{T}$, it remains computationally tractable for realistic instances. Further, both constraints sets (76) and (77) consist of knapsack constraints that are well researched (Cacchiani et al., 2022). Consequently, the overall computational effort primarily depends on the estimation of $Q(\gamma^{LT}, \gamma^{ST})$.

Estimating $Q(\gamma^{LT}, \gamma^{ST})$ is itself an optimization problem where the goal is to minimize the cost of performing maintenance at a set of OWFs during a planning horizon $\mathcal{T}^P$ with a given fleet allocated to a set of O&M bases. This corresponds to modeling the day-to-day operations of the fleet. Ideally, this model should be a multi-period version of the WFMRSPP with multiple OWFs and O&M bases to give the most precise estimate. However, as the WFMRSPP is in itself difficult to solve in a reasonable amount of time, the WFMSP may instead serve to estimate $Q(\gamma^{LT}, \gamma^{ST})$. Most papers use a middle ground between the two: As the focus is on decisions at the strategic or tactical level, the short distances between OWTs within the OWF are negligible, and to ensure computational tractability, only the longer distances between the OWFs and O&M bases are considered.

A common feature of papers with a tactical perspective is that the problem of estimating $Q(\gamma^{LT}, \gamma^{ST})$ is modeled as a (simplified) variant of the WFMRSPP with multiple time periods. Halvorsen-Weare et al. (2013) and Gundegjerde et al. (2015) use *activity bundles*, which represent sets of activities that can be performed in parallel by a given vessel, and introduce decision variables indicating whether the vessel performs the activities of each bundle in a given time period.

Inspired by cutting patterns for the cutting stock problem, Stålthane et al. (2019) extend the notion of bundles to *maintenance patterns*. Similarly to the Dantzig-Wolfe decomposition of the WFMRSPP, as described in Section 5.1.2, a maintenance pattern can be thought of as a path through a simplified network, where each node represents a type of MT (rather than an actual MT), and all arc parameters are assigned average values. Using these maintenance patterns, they formulate a generalized set partitioning model to estimate $Q(\gamma^{LT}, \gamma^{ST})$, where the model selects one pattern for each vessel in each time period with the goal of covering all maintenance tasks.

Gutierrez-Alcoba et al. (2019) extend the maintenance patterns by including the number of technicians each pattern requires, and use this to determine the number of technicians needed at the O&M bases. Another detail that the tactical papers consider is that some vessels, such as jack-up and accommodation vessels, can stay offshore for several periods. The articles with a strategic perspective, on the other hand, consider the possibility of chartering out or selling vessels, thereby generating revenue as the fleet is adjusted depending on the maintenance demand.

In this basic version of the WFMFSMP model presented in this paper, it is assumed that all vessels and O&M bases can be freely selected. This might hold for the strategic variant of the problem, but

for the tactical variant, some vessels and O&M bases may already be selected prior to the time horizon of the model and therefore given as input to the model. This occurs when the WFMFSMP is used to adjust the fleet during the lifetime of the OWFs. Furthermore, the presented WFMFSMP model does not distinguish between onshore and offshore O&M bases. However, two papers investigate the deployment of mother vessels, which is unique to OWFs. These vessels serve as offshore docking stations for smaller daughter vessels and can accommodate technicians. Stålhane et al. (2016) treat mother vessels as offshore O&M bases, however, they can also be employed themselves for MTs, which is investigated in Stålhane et al. (2019). They model this by letting the mother vessel be both an offshore O&M base as well as a vessel assigned to this base together with the daughter vessels, with the accumulated cost of all the vessels assigned to the offshore O&M base.

A special case of the WFMFSMP is the offshore *WF maintenance fleet size problem* (WFMFSP), which is restricted to considering a homogeneous fleet of vessels and the use of O&M bases is omitted. In general, this makes the WFMFSP less complex than the WFMFSMP. The WFMFSP has received limited attention in the literature. However, studies on the WFMFSMP reveal that some highly specialized vessel types, such as jack-up vessels (and other HLVs), act as cost drivers in the WFMFSMP (Dalgic et al., 2015a). The high level of specialization means that these types of vessels have little or no overlap in usage with other vessel types, which motivates looking at these separately from the rest of the fleet. Stålhane et al. (2017) present a model for the jack-up vessel chartering problem and propose to solve it with sample average approximation (SAA). Simulation has also been applied to study the same problem (Dalgic et al., 2015c), the related problem of resource sharing of jack-up vessels between multiple OWF operators (uit het Broek et al., 2019), and the chartering of mother vessels (Dalgic et al., 2015b).

Another interesting special case of the WFMFSMP studied by Schrottenboer et al. (2020) arises when the fleet is given and the aim is to assign it to O&M bases on a tactical level such that a set of OWFs can be maintained at an operational level fulfilling some service requirements. This problem occurs when the service provider is not the same as the OWF operator. For instance, it is common in the industry that the OWF operator has a full-service contract for the first 2–5 years of the OWFs’ life, where the OWT manufacturer handles all maintenance while guaranteeing a certain level of availability.

When it comes to solution methods, the first five papers use a commercial MIP solver to solve the problem; however, the methods used to pre-generate activity bundles and maintenance patterns are not exhaustive, so we prefer to classify them as matheuristics. A labeling heuristic then serves as a part of a matheuristic by generating feasible activity bundles. Although they do not formulate the decomposed model explicitly, Gutierrez-Alcoba et al. (2019) rely on the same solution approach: a promising subset of columns (i.e. activity bundles) that constitute solutions to their second-stage problem are pre-generated, and the overall problem is solved using a default MIP solver. Stålhane et al. (2021) and Bolstad et al. (2022) use an integer L-shaped method and a greedy randomized adaptive search procedure (GRASP), respectively, to solve their dual-level stochastic model.

5.3. Handling uncertainty in the operations and maintenance phase

As is the case for the planning of OWF installation, that also requires maritime logistic operations, weather conditions in terms of wind speed and wave height governing the vessel operability is one of the main sources of uncertainty when planning maintenance at OWFs. Another main source of uncertainty in maintenance planning is the occurrence of failures, which create a demand for maintenance.

The short time horizon in the operational problems - and the fact that the WFMRSMP is already difficult to solve - is probably the reason that most of the papers on the operational problems treat weather conditions and failures as known. However, Besnard et al. (2011) extend the WFMSP formulation of Besnard et al. (2009) to a two-stage stochastic model, introducing uncertainty in the weather conditions.

When it comes to the WFMFSMP, two papers consider the weather conditions as uncertain. Irawan et al. (2021) capture it in a two-stage stochastic optimization model where it affects the travel time, transfer time of technicians and equipment, and maintenance time, while Feng et al. (2024) utilize distributionally robust optimization to handle uncertainty in prediction errors of OWTs’ maximum available active power outputs, and the travel time of maintenance vessels. In addition, Irawan et al. (2022a) and Si et al. (2025) circumvent direct handling of uncertainty by splitting the WFMSRP into the MSP and the MRP as previously described.

The focus on long-term decisions in the WFMFSMP makes it natural to take uncertainty into account when solving the problem. All but the first paper listed in Table 6 consider operational uncertainty in weather conditions and failures, and capture this uncertainty through two-stage stochastic models where the first stage is analogous to the WFMFSMP, and the second stage estimates $Q(\gamma^{LT}, \gamma^{ST})$. Depending on whether the WFMSP or the time expanded WFMRSMP is used to model $Q(\gamma^{LT}, \gamma^{ST})$, the uncertainty in vessel operability due to weather conditions can be captured by making the parameters A_{vt} or T_{vt}^{MAX} stochastic. In addition, the penalty cost C_i^P is a stochastic parameter since the (lost) electricity production depends on the wind speed. The uncertainty in failures may be captured by making the parameter C_{it}^D stochastic, as the downtime cost is a function of the number of time periods between a failure occurs and maintenance is performed. To avoid preemptive maintenance, the downtime cost can be set to infinity for time periods before the failure occurs.

Within the literature taking a tactical perspective, Gundegjerde et al. (2015) in addition consider tactical uncertainty in electricity prices and vessel spot rates between the first stage and the second stage of their three-stage stochastic model. Within the literature having a strategic perspective, Stålhane et al. (2021) and Bolstad et al. (2022) capture strategic uncertainty in electricity prices and stepwise OWF development, in addition to the operational uncertainty in weather conditions and failures, through a *dual-level stochastic model* (see e.g., Kaut et al. (2013) and Pantuso et al. (2015)). This modeling method enables uncertainty on two different time scales under the assumption that the long-term uncertainty is independent of the short-term uncertainty.

5.4. Data instances

As for the installation data sets, most of the surveyed papers on O&M base their data instances on synthetic data, however, data sets based on real OWFs are used by: Irawan et al. (2021) (*Thanet*) Feng et al. (2024) (*Guangdong Shanwei*), Ma et al. (2024) (*Dalian Zhuanghe*), and Si et al. (2025) (*Empire*).

The reference case presented by Dinwoodie et al. (2015) serves as a starting point for the majority of articles surveyed in this section (Stålhane et al., 2015; Raknes et al., 2017; Stålhane et al., 2019; Schrottenboer et al., 2020; Stålhane et al., 2021; Bolstad et al., 2022). It consists of a single OWF with 80 OWTs located 50 km from the O&M base. Weather data (wave heights and wind speeds) are sourced from the *FINO Databank* (BSH, 2025), while the WT power curves are based on the collection gathered by Staffell (2012), which contains power curves for WTs ranging from 0.4MW to 6MW.

Three different vessel types are presented in Dinwoodie et al. (2015): CTVs, *field support vessels*, and HLV. For each of these, details on capabilities are provided in terms of governing weather criteria – wave height (and wind speed for the HLV), mobilization time and cost, sailing speed, technician capacity, charter rate, and maximum time offshore. Five failure categories are included. That is, *manual reset*, *minor repair*, *medium repair*, *major repair*, and *major replacement*. For each category, the repair time, the number of technicians required, the type of vessel required, repair cost, and the failure rate are presented. In addition to the five types of failures, similar information is provided for *annual service*.

Stålhane et al. (2019) expand on the data set of Dinwoodie et al. (2015) by adding information from Sperstad et al. (2017) on four additional vessel types: *surface effect ship*, *small accommodation vessel*, *mini mother vessel*, and *daughter vessel*. Furthermore, Schrottenboer et al. (2020) have developed a set of 54 benchmark instances differing in the number of OWFs (1-3 OWFs), the number of OWTs in each OWF(s) (30-90 OWTs), and number of vessel lease term periods (1-3 periods) using the previous data of Dinwoodie et al. (2015), Stålhane et al. (2019), and Gundegjerde et al. (2015) as well as data gathered from industry partners.

In addition to the data sets listed above, a number of papers (Schrottenboer et al., 2018, 2019; O’Neil et al., 2023) base their studies on the data set presented in Dai et al. (2015) and its extension by Irawan et al. (2017a). Another set of papers (Irawan et al., 2021, 2022a; O’Neil et al., 2023) use a data set developed in the EU funded project LEANWIND, which is based on the *West Gabbard* OWF, with OWT characteristics from Desmond et al. (2016), and Irawan et al. (2017a) as a source for the remaining input data.

6. End of life

When mentioning the expected lifetime of 25–30 years for the OWFs in the introduction to this survey, we referred to what is known as the *design life*. As the name indicates, this is the lifetime used in the

structural strength verification when designing an OWT. The *operating life*, on the other hand, is the actual lifetime of an OWT (Natarajan et al., 2020). As the final phase of the OWF is approaching, the potential *end of life* (EOL) strategies can be split into three categories (Shafiee and Animah, 2017; Luengo and Kolios, 2015). *Life extension* is the prolonging of the operational life of the OWTs, and typically involves a major service inspection to determine which maintenance actions are required. Often the tower is still functional, while other components such as the generator and blades might need replacement or major repair.

Repowering consists of reusing some parts of the original OWF to establish a new generation of wind turbines in the wind farm. The reused parts are often those that are overdimensioned for the first generation of the wind farm to guarantee they last the full life cycle, e.g., the foundations. *Decommissioning* is the full dismantling of the wind farm in accordance with international regulations (among them the UN Law of the Sea (UNCLOS)) (Adedipe and Shafiee, 2021).

Motivated by the fact that many of the onshore WFs in Europe are aging, Ziegler et al. (2018) did a review to study research needs related to life extension. Their findings suggest that a significant market for EOL solutions will develop over the next few years. So far, only a few OWFs have reached their EOL phase. However, with the number of OWTs already installed and projected to be installed in the coming years, there is reason to believe that over time a similar market will evolve for the offshore wind industry. EOL strategies for OWFs have not received much attention in the literature, although several studies point out the need for research on efficient logistics planning in relation to EOL strategies (Ortegon et al., 2013; Topham and McMillan, 2017; Eligüzel and Özceylan, 2022). We found only one article focusing on decision support for which EOL strategy to pursue: Jadali et al. (2021) review literature related to EOL and uses this to develop a multi-attribute framework guiding decision makers on which strategy to choose.

6.1. Life extension

As there are generally no subsidies for operating WFs beyond their design life, the income is driven by the price of electricity in the uncertain spot market (Nielsen and Sørensen, 2021; Walgern et al., 2023). Hence, this calls for good decision support to evaluate whether life extension is worthwhile. Nielsen et al. (2019) formulates a Bayesian decision problem capturing decisions regarding which analysis and inspections to perform before making a final decision on whether or not to extend the lifetime of an OWF.

Niknam et al. (2015), on the other hand, investigate how information on component degradation obtained from prognosis models at the end of the OWTs’ design life can be exploited through multi-objective optimization to maximize both the energy production and the operational life of the OWTs, while taking maintenance into account.

The possibility of life extension is closely related to how O&M was carried out throughout the OWT’s design life. This connection has started to receive attention in the literature related to OWF control. For example, Del Pozo González and Domínguez-García (2022) show how the fatigue load can be more evenly distributed between the upstream OWTs and the downstream OWTs while satisfying the required energy production, and Harrison et al. (2020) find that the fatigue load on the blades can be reduced while simultaneously increasing the energy production. When it comes to the connection with maintenance planning, there is, to the best of our knowledge, a gap in the literature that has yet to be explored.

6.2. Repowering

In 2018, the 20-year-old Bockstigen OWF in Sweden, consisting of five OWTs, was the first OWF to be repowered. The original towers, foundations, and cables were reused, while blades, nacelle, and control systems were replaced with components from refurbished OWTs. The now repowered OWF is expected to have an operational lifetime of 15 years (Gredes, 2018). Despite the fact that repowering is already taking place, the literature addressing optimization problems related to the topic is sparse.

Hou et al. (2016) and Hou et al. (2017a) both look into how the original foundations in an OWF can be reused for a second generation of OWTs. The idea of Hou et al. (2016) is to reuse the existing foundations for a new generation of smaller OWTs, as well as constructing an additional set of OWTs in the OWF to meet the same overall capacity as the original OWF. The problem is formulated as a non-linear optimization model and solved heuristically with adaptive PSO. Hou et al. (2017a) assume that the old OWTs will be dismantled over a period of time such that there are two replacement periods

with four years between them. The problem consists of selecting which of the old OWTs that should be replaced, and by which type(s) of new OWT, in each period, while respecting the capacity of the cables. They formulated this as a non-convex optimization problem that minimizes the LCOE by taking wake loss of the OWTs into account and solved it with adaptive PSO.

The underlying assumption of Hou et al. (2016) is that all OWTs are in the same health condition after 20 years of operation, while Hou et al. (2017a) assume that OWTs on the perimeter of the OWF and OWTs facing the prevailing wind direction are in a poorer health state. Liu et al. (2022), on the other hand, propose a repowering strategy based on a health evaluation. It has two steps: i) the health condition of each WT is evaluated using dynamic deterioration, and ii) an optimization problem is solved similar to the one in Hou et al. (2017a) using a genetic algorithm (GA).

6.3. Decommissioning

Despite the fact that it is mandatory for OWF projects to have a decommissioning plan prior to project approval, decision support for this problem has received little attention in the literature. This is probably related to the juvenile state of most wind projects, where the main focus is on setting up OWFs. However, since the decommissioning costs are part of the overall cost estimate for the entire wind project, it is crucial to have a decommissioning plan that is sufficiently accurate from the beginning. Currently, published decommissioning plans are often optimistic in terms of time and cost (Topham and McMillan, 2017). Furthermore, Adedipe and Shafiee (2021) found that logistics and planning were among the activities that made up the largest proportion of the total cost. Altogether, this calls for more research focusing on the detailed modeling of the decommissioning process.

Decommissioning is often described as the reverse process of installation. However, there are some characteristics distinguishing the two problems: Where the overall goal of the installation problem is to balance cost and completion time in order to minimize both, the goal of the decommissioning problem is cost minimization. Furthermore, the installation problem involves a flow of components from one point, being the port, to many points at the OWF, while the decommissioning problem involves a flow of used components from many points at the OWF to several potential ports. The problem shares similarities with the offshore plug and abandonment campaign planning problem for decommissioning of oil wells (see e.g., Bakker et al. (2019)), but does not involve specialized handling of the abandoned structures such as the foundations, which is necessary in the case of oil wells to prevent pollution of the surrounding marine environment (Vrålstad et al., 2019).

Irawan et al. (2019) were the first to formulate a model for the decommissioning of an OWF. The goal is to minimize the overall costs of vessel chartering, inventory, processing, and land-based transportation by considering the schedule of jack-up and barge vessels, as well as the flow of components from the OWF to the recycling or landfill area. The authors present a matheuristic where finding compact schedules for the jack-up vessels is central, as the vessel chartering costs are the main cost driver in the objective function.

We also found a few simulation studies. Beinke et al. (2018) looked into how tactical decisions regarding vessel chartering affect the cost of decommissioning when the deployment of the vessels is affected by uncertainty in weather conditions. McAuliffe et al. (2019) test the sensitivity of a decommissioning plan for an OWF when varying the available vessels and technicians, operational weather restrictions, operation duration, and distance from port to OWF, while costs and weather conditions are uncertain.

7. Future trends and research directions

The offshore wind energy industry is undergoing rapid development. In the coming years, the installed offshore wind capacity is expected to grow exponentially (GWEC, 2024). Although this progress is positive for the green energy transition, it also creates challenges in terms of bottlenecks in the supply chain limiting e.g., the number of technicians, vessels, and components available in the market (GWEC, 2023). Therefore, there is an increased need for better resource planning throughout the OWF life cycle to further decrease the LCOE and ensure the viability of offshore wind energy.

Furthermore, new technological advancements such as floating OWTs and offshore vessel charging stations, introduce new planning problems to the field. This gives rise to exciting new model applications. Overall, this creates a great variety of interesting research opportunities contributing to the development

of offshore wind energy. In this section, we point out three main directions: i) maturing the field by standardizing data sets and models to improve solution methods, ii) development of holistic approaches to include more aspects in the models, and iii) new applications driven by technological advancements. We consider them in Section 7.1, Section 7.2, and Section 7.3, respectively.

7.1. *Maturing the field*

Currently, we see an uneven distribution in the attention that problems within each phase of the life cycle have received. OWF design in phase II has received by far the most attention, which is reflected in the maturity level of the solution methods. To respond to supply chain bottlenecks and decrease the LCOE, it is of great importance to mature the planning problems within the subsequent life cycle phases, as well as standardizing the mathematical models and improving the solution methods.

An integral step to make the research field more accessible to new researchers, and enable fair comparisons between solution methods, is the development of standardized data sets. Although there exist some data sets, what remains to be developed are more comprehensive data sets based on current data, as much of the existing data on, e.g., OWTs and vessels, is not up to date with the rapid technological advancements of recent years.

Such standardized data sets should ideally contain data for several different OWF sites (e.g., weather conditions, seabed map and distance to shore), modern OWT types (power curve) including their components (e.g., failure rates and cable throughput), different vessel types including new concepts such as mother vessels and vessels aimed at floating wind (e.g., operability in different weather conditions, technician and component capacities, etc.), and technicians (e.g., skill level, shift length, etc.).

An integral step to make the research field more accessible to new researchers, and to allow fair comparisons between solution methods, is the development of standardized data sets. Although some data sets exist, what remains to be developed are more comprehensive sets based on recent data. Such standardized data sets should ideally include data for several different OWF sites (e.g., weather conditions, seabed maps, and distance to shore), modern OWT types (power curves) including their components (e.g., failure rates and cable throughput), different vessel types including new concepts such as mother vessels and floating wind-specific vessels (e.g., operability in different weather conditions, technician and component capacities), and technicians (e.g., skill level, shift length).

7.2. *More holistic approaches*

In the surveyed literature, we have seen multiple extensions of the basic models to include additional aspects, e.g., the integration of OWF layout and cable routing (e.g., Fischetti and Fischetti (2023) and Butterwick et al. (2023)) in OWF design (Phase II), and technician skills and allocation (e.g., Irawan et al. (2022a) and Schrottenboer et al. (2018)) in O&M (Phase IV). It is natural to continue this development as the research field matures. One interesting extension is to integrate supply chain optimization (Irawan et al., 2018) with scheduling of installation vessels in Phase III.

Whereas this is an example of obtaining more holistic approaches by including more aspects within each life cycle phase, models spanning more phases of the life cycle might further reduce costs. A relevant problem to consider in this context could be to connect phase V, end of life of one OWF, with phase III installation of a new OWF on the same site into an integrated problem. Another possible integration could be between phase IV O&M and phase V end of life. As the condition of the OWTs depend on the maintenance carried out throughout the lifetime, it could be interesting to connect maintenance planning with end of life to evaluate the level of maintenance in connection with the end of life strategies.

When resources become increasingly scarce, an interesting topic is to investigate how these can be shared among OWFs. Currently, each OWF has its own substation and cable infrastructure. Large savings could probably be achieved if OWFs that are close to each other share some part of the infrastructure. This could also make the supply more robust, by enabling alternative transmission lines and adding flexibility to the system, since surplus electricity in one region can be transferred to other regions. One way to obtain this is by establishing *energy islands* that serve as a hub for multiple OWFs and provide facilities to transmit electricity to several regions. Furthermore, electricity can be transformed at the energy islands into other types of energy and stored for later use (Ramboll, 2023).

Another interesting research topic is the sharing of maintenance vessels between OWFs. Currently, each wind farm operator and manufacturer usually considers only OWFs within their own portfolio.

However, there might be potential for cutting costs by exploring different vessel sharing policies, e.g., *partial sharing* and *full sharing* to find a balance point between flexibility and vessel chartering cost. In this context, it is essential to incorporate *fairness criteria* into the objective function to ensure that the vessels are shared fairly between the OWFs.

As a consequence of the direct link between the harsh maritime environment and the relatively higher LCOE for offshore wind (compared to onshore wind), it is standard to incorporate uncertainty in weather conditions in models within the installation and O&M phases. In addition, O&M models include uncertainty in failure rates. Some models explore other types of uncertainty, e.g., electricity prices, vessel spot rates, and the completion of each stage of an OWF's development project (as large OWFs are often developed and installed in several stages). Although wind farm projects are currently subsidized in most markets to aid the acceleration of the green energy transition, e.g., by guaranteeing a minimum price for the electricity produced, this may not be the case in the future when the industry is more mature. Therefore, it could also be worthwhile to continue exploring the incorporation of uncertainty in electricity prices.

Future studies could also account for the uncertainty in lead time and (spot) charter rates for vessels and components, as these resources become increasingly scarce. As a consequence, the uncertainty of when each stage of an OWF is fully developed increases and must be taken into account. The time of decommissioning is another uncertain parameter, as it might be possible to extend the lifetime beyond the design life. However, an OWF may also be decommissioned earlier, e.g., the OWF Yttre Stengrund was dismantled after just 15 years of operation due to difficulties in sourcing spare parts (POWER, 2016). This is a direct consequence of the rapid development of new and larger OWT models, which means that OWTs (especially rare ones) and their spare parts are outdated just as fast.

7.3. New applications driven by technological advancements

In the following, we will address a number of technological advancements and discuss how they may lead to new and interesting OR problems.

Upscaling of OWTs: Currently, we see *larger OWTs* being designed and installed every year. Although this decreases the LCOE as the costs are averaged over greater energy outputs, it also complicates cost reductions. The larger design requires development of new vessels and methods for installation, maintenance, and eventually decommissioning. Furthermore, the structural design of OWTs and their long-term behavior are less understood. This introduces uncertainty – hence financial risk – and may lead to a higher number of breakdowns for these new OWTs (Modern Power Systems, 2023). It remains an open question whether continuous upscaling is profitable compared to increased familiarity with the OWTs that can bring benefits in terms of better behavior prediction (e.g., on failures), and use of standardized models for installation, maintenance, and decommissioning. Mathematical models that can aid the analysis of trade-offs between continuous upscaling and standardization of logistic operations would be a valuable contribution to this discussion.

Digitalization: At present, the wind energy sector is undergoing a digital transformation. By gathering sensor data in digital twins to represent the OWFs, it is possible to monitor the OWFs more precisely. It also paves the way for more sophisticated use of data, e.g., for prognostics on power output and component degradation (WindEurope, 2018). This development not only enables advanced prognostics but also opens the door for research where machine learning and OR are integrated with each other with the aim of enhancing decision-making under uncertainty (see, e.g., Bengio et al. (2021); Gambella et al. (2021) for surveys on this integration). One interesting application is to incorporate prognosis on component degradation into maintenance planning. Ideally, an OWT is maintained the day before it fails. Then the full lifetime of the failed component has been utilized, and the OWT is not exposed to unnecessary downtime. As failure predictions become better, it is interesting to investigate how big a role the precision of failure information plays. For example, by using the existing literature on maintenance planning (Phase IV), e.g., fleet size and mix, as a starting point, different maintenance strategies such as *condition-based maintenance* and *predictive maintenance* could be compared. In a condition-based maintenance strategy, a component is maintained when it reaches a certain degradation threshold, whereas in a predictive maintenance strategy, predictions on the *remaining useful lifetime* (RUL) until failure are used to determine when a component is maintained.

Floating OWTs: With many near-coastal sites already occupied, the OWFs are moving to deeper waters (IRENA, 2025), requiring floating foundations as the increased water depths make bottom-fixed foundations unsuitable (IRENA, 2023). The anchors for floating foundations, and longer distance to shore, may lead to more advanced cable layout problems. Further, new strategies such as towing of the OWTs to/from designated ports with sufficient water depth, and self-elevating cranes, are developed for installation (Hong et al., 2024) and maintenance (McLean, 2025). Developing appropriate mathematical models to capture these challenges, and solving them could improve the scheduling of operations and aid the evaluation of different strategies.

Vertical axis wind turbines (VAWTs): So far the HAWT design (see Figure 2a) has been predominant. However, recent studies suggest that VAWT (see Figure 2b) might be a suitable candidate for floating OWFs. Since the nacelle is placed at the bottom, the center of gravity is lower. Therefore, they require smaller foundations and are easier to operate and maintain (Ghigo et al., 2024). VAWT turbines have an asymmetric wake profile, and may even increase energy production if they are located close to each other (Dabiri, 2011), but proper wake models that are easy to calculate, and that have been validated in practice, are still missing. As research on technical models progresses, the formulation of optimization models complementing these is essential to assess strategic choices—such as whether VAWTs outperform HAWTs on floating foundations—when evaluating across the life cycle.

Power-to-X: Technologies for transforming electricity to other energy forms such that it can be stored for later use are collectively known under the term *power-to-X* (PtX). One of the suggested PtX technologies is *hydrogen-producing OWTs*, where the electricity is immediately transformed to hydrogen through electrolysis, making it possible to replace the transmission grid’s from cables with tubes. Since the material consumption of tubes only grows moderately with the throughput, this could be a cheaper way of transmitting large amounts of energy (Patonia et al., 2023).

Another PtX solution is to use batteries, which allow for a better utilization of the transmission grid, since peak production can be stored locally. As energy production often fluctuates during a day, being able to store a few hours of production could reduce investments in cable connections, and increase yield by transmitting (and selling) the power at low-production times of the day (Colombo et al., 2024). An interesting extension of the cable routing problem (Phase II) arises in this context as the PtX solutions affect decisions on network design and the capacity of cables and/or tubes.

Offshore charging stations: Recently, the idea of using OWFs as offshore charging stations for E-vessels has been introduced (ScottishPower Renewables and Stillstrom by Maersk, 2025). This could be *electric SOVs* performing maintenance, however, it also has a broader potential, e.g., for fishing or cargo vessels. Interesting research questions could arise from this, e.g., in the industry intersection between wind energy and shipping, the location of charging stations at OWFs in relation to shipping routes could be investigated.

8. Concluding remarks

The offshore wind energy industry is young and undergoing rapid development. This is positive for the green energy transition, but it also introduces challenges such as bottlenecks in the supply chain and integration of new technologies. Therefore, efficient resource planning within each phase of the offshore wind farm life cycle is critical to reduce costs and make offshore wind energy a viable alternative. Operations research has the potential to contribute significantly to this goal.

In this survey, we have provided a detailed overview of the existing literature and presented standard mixed-integer programming models for selected problems within the five phases of the life cycle: area selection, offshore wind farm design, installation, operations and maintenance, and end-of-life. The literature within each phase has a different level of maturity, which is reflected in our discussion of problem definitions, models, and solution approaches. The offshore wind farm design phase stands out as the most developed, characterized by standardized problem classes and a wide range of solution approaches. In comparison, the installation and the operations and maintenance phases are less mature, with current research efforts primarily aimed at introducing richer and more realistic problem formulations. For the area selection and end-of-life phases, which are less explored in the literature, we have briefly discussed possible problem settings.

We have also highlighted future trends within offshore wind and pointed to potential research directions. There are ample avenues to explore along three axes: (i) maturing the field by standardizing models and datasets (particularly for the later phases), (ii) developing more holistic approaches by integrating additional aspects into current models, and (iii) formulating new models to provide decision support for emerging problems arising from technological advancements in the industry. We aim for this survey to serve as a foundation for future research and practical advancements in this exciting and rapidly evolving field.

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